OBLIGATORY TRIPLE SYSTEMS: CANONICAL ATOMS AND EXACT FINITE SPECTRA

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11 October 2026

Abstract. Erdős Problem 593 asks which finite triple systems occur in every triple system of uncountable chromatic number. We give an independent proof of the classification and derive its finite structural consequences. After isolated vertices are removed, a system is obligatory if and only if it is linear, every hyperedge-node of its Levi graph is incident with a bridge, and every Berge cycle has even length. Its canonical atoms are single triples and private-vertex expansions of finite 2-connected bipartite graphs. For reduced obligatory systems with at least one hyperedge, we determine the exact parameter and atom-count spectra, including the parity obstruction at cycle rank one. Supported one-point factorizations form a product of partition lattices. Every attachment excess profile can be realized while keeping the atoms and component count fixed, and the lattice type recovers the profile. For connected systems of positive cycle rank, an exact budget relates cyclic-rank splitting and excess core order to the available factorization rank. At every prescribed original-edge piece count, sharp binomial and Stirling-number bounds describe the number of factorizations. Binary joints and a common joint characterize the interior equality cases; the extremal assemblies can have the same atoms and global parameters. These results distinguish the constraints of cyclic content from the freedom of assembly. For the associated graph cores, we also prove exact edge-marginal entropy bounds for uniform Potts laws and nearby general pair laws, and common-power density bounds for two-level regular hosts. For a class of weighted signed projection lifts, a comparison retains the full coarse density and gives an explicit positive cycle margin for every choice of the six positive exponents. These comparisons are separate from the obligatory classification, and their unrestricted versions remain open. The classification and specified structural extensions are verified in Lean 4. Profile realization, full lattice spectra, the new counting endpoints and the entropy–density arguments are not yet fully formalized. All arguments are in ZFC.

2020 Mathematics Subject Classification. Primary 05C65; Secondary 05C15, 05C35, 05C40, 05C63, 06A07, 03E05.

Keywords. Erdős Problem 593, obligatory triple systems, uncountable chromatic number, canonical decomposition, partition lattices, entropy, homomorphism densities, Lean 4.

Introduction

Although the host systems are infinite, the classification is determined by finite structure. A finite triple system is *obligatory* if it occurs in every triple system of uncountable chromatic number. Erdős asked for a characterisation of

these finite systems [1]; the question is catalogued as Erdős Problem 593 [2]. We give both constructive and intrinsic descriptions of the answer.

For a finite graph J, write J^+ for its private-vertex expansion. Let $\mathcal{B}$ be the smallest class containing J^+ for every finite bipartite graph J and every finite edgeless triple system, and closed under finite disjoint unions and one-point amalgamations. Membership in $\mathcal{B}$ is understood up to triple-system isomorphism. If F is a triple system, let F° denote the system obtained by deleting its isolated vertices.

Theorem A. *For every finite triple system F, the following are equivalent:*

(1) F is obligatory;
(2) $F \in \mathcal{B}$;
(3) F° is linear, every hyperedge-node of $I(F^\circ)$ is incident with a bridge, and every Berge cycle of F° has even length.

The classification first appeared in Li's preprint, posted on 23 June 2026 [3, Theorem 1.1]. The present proof and Lean implementation were developed independently. Both proofs use one-apex sequence lifts and Levi bridges, but their finite trace arguments differ: Li uses a bridge-trace formulation and graph derivatives [3, Sections 3–4 and Theorem 4.6], whereas we decompose traces into expansion fibres and a support-incidence forest. We cite Li's work for chronology and comparison; none of its results is used below as a black box.

Finite structure beyond the classification. Theorem A answers an embedding question about infinite hosts. The further results concern a different question: how much finite structure is fixed by cyclic content, and how much remains free under one-point assembly? The finite systems considered in this overview are reduced (they have no isolated vertices), obligatory, and have at least one hyperedge. Write $n = |V(F)|$, $m = |E(F)|$, and c for their number of components; put $s = n - m$ and $\beta = 2m - n + c$, the Levi cycle rank. The canonical normal form identifies the atoms as single triples and expansions of finite 2-connected bipartite graphs (Theorem 5.3). Block decomposition itself is classical [4, 5]; the point here is the exact list of allowable block types and what it implies about the systems assembled from them.

The finite results address two complementary classification questions. The numerical spectra determine which global parameters and canonical atom counts are possible. The local partition product describes the decompositions available once the atoms have been fixed. These questions are distinct: assemblies with identical atoms, order, size and cycle rank can still have different factorization lattices and different numbers of decompositions.

The numerical consequences are exact existence statements, not only bounds. Theorem 10.3 determines every feasible order, size and component count. Theorem 10.9 and Proposition 10.12 then determine all possible canonical atom counts k at fixed s, β, c, including the parity obstruction when $\beta = 1$. For connected systems, the indecomposability phase diagram separates parameter

ranges where every system is decomposable, where both behaviours occur, and where every system is indecomposable (Theorem 10.10). Its boundary cases identify cycle, even-theta and complete-bipartite expansions, and the extremal decomposable assembly (Corollary 10.11). These conclusions do not say that numerical parameters determine every core up to isomorphism.

The remaining freedom is described by a lattice, not another numerical invariant. At each point p shared by $\mu(p)$ canonical atoms, one may choose a partition of those atoms independently. The supported one-point forest decompositions of the original hyperedge set, ordered by refinement, therefore form $\mathcal{D}(F) \cong \prod_{p \in S(F)} \Pi_{\mu(p)}$, where $S(F)$ is the set of shared points and Π_r is the partition lattice on r elements (Lemma 10.15). Its rank is $N = k - c$. Every integer partition $\lambda \vdash N$ is realized as an attachment excess profile by reassembling the same atoms with the same component count, order, size and cycle rank (Lemma 10.16). This gives the exact factorization-lattice spectrum: its types are $\prod_i \Pi_{\lambda_i+1}$, and the classical characteristic polynomials of partition lattices [6, Chapter 3] recover λ (Corollary 10.17). The classification is of lattice types, not of all triple systems with the same parameters.

Cyclic content and assembly freedom are linked by an exact budget. For a connected system of positive cycle rank, the lattice rank $N = k - 1$, the cost of distributing positive rank among several atoms, and the total core-order excess above the fixed-rank minima sum to $s - 2 - \lceil 2\sqrt{\beta} \rceil$ (Proposition 10.18). Thus rank splitting and excess core order reduce the available decomposition rank. Zero cost means that one minimum-order cyclic atom carries all positive rank and the remaining atoms are single triples; it does not imply a unique isomorphism type for the cyclic core.

Assembly geometry also controls the number of decompositions at each prescribed piece count. If $d_j(F)$ counts decompositions into exactly $k - j$ nonempty original-edge pieces, then $\binom{N}{j} \le d_j(F) \le S(N+1, N+1-j)$ for $0 \le j \le N$, where $S(a, b)$ is a second-kind Stirling number [7]. For $N \ge 2$, equality at any interior j characterizes binary joints at the lower bound and a single common joint at the upper bound (Corollary 10.19). Both extremal profiles can be attained with the atom list and component count fixed. The example following that corollary has identical atoms and global parameters but four or five decomposition choices. These counting consequences quantify the freedom left by the canonical structure; they are not assertions of new general Stirling-number identities.

Quantitative questions suggested by the atoms. The description of the cyclic atoms also identifies graph cores for finite quantitative questions. Every simple bipartite subdivision of K_4 is 2-connected and has cycle rank three, so its private-vertex expansion is an allowable atom. This observation motivates, but does not prove, a density comparison for the original graph. Section 11 studies exact pair-marginal entropy witnesses and regular-host homomorphism densities for these cores. The first results cover uniform q-state Potts tables,

arbitrary tables in specified open families, and hosts with two centered spectral levels. A further transfer theorem allows signed projection channels over mildly nonuniform original measures, with no bound on the number of spectral levels (Proposition 11.8). They are not a universal Sidorenko theorem.

The entropy formulation has established predecessors [8, 9]; local density comparisons near independence are also classical [10]. The positive-spectrum comparison has its own prior literature [11]. We give the finite arguments needed for our restricted statements and separate them from the two unrestricted questions. Neither the graph comparisons nor their novelty is inferred from the classification of obligatory triple systems. Some of these core graphs already belong to known scalar-density classes: graphs with a vertex complete to the opposite part are covered in Conlon et al. [12]. Such coverage is not a new graph family here. More recently, Im et al. [13] proved subdivision and even-theta substitution results. Their uniform substitution theorem and their nonuniform theorem with path-count divisibility hypotheses do not settle arbitrary choices of the six subdivision lengths considered below.

The classification and substantial structural extensions have accepted Lean implementations. The manuscript also proves results whose full formal interfaces remain unfinished: profile realization, lattice spectra, maximal-chain and height endpoints, and the new counting statements. Thus a manuscript proof is not, by itself, a claim that the corresponding Lean interface has been exported and accepted. The entropy and density arguments have no accepted Lean implementation. Appendix A records the implementation routes, validation records and remaining obligations.

Context and proof strategy. The private-vertex expansion J^+ also appears in finite extremal hypergraph theory. Kostochka, Mubayi and Verstraëte study its Turán extremal function [14]. Here the same operation supplies the canonical cyclic pieces, but the host condition is uncountable chromatic number rather than edge density.

The graph analogue was proved by Erdős and Hajnal: the obligatory finite graphs are exactly the bipartite graphs [15, Corollary 5.6 and Theorem 7.4]. The classical nonlinearity obstruction for uniform hypergraphs is due to Erdős, Hajnal, and Rothschild [16, Theorem 2, p. 532]. Obligatory triple systems were studied further by Erdős et al. [17], Komjáth [18], Hajnal and Komjáth [19], and Komjáth [20]; the expansion theorem of Reiher [21, Theorem 1.2] supplies the positive atoms used below.

In the finite setting, arbitrarily large chromatic number is compatible with arbitrarily large girth: Erdős proved this for graphs [22, pp. 35–37], and Erdős and Lovász established a uniform-hypergraph counterpart [23, Theorem 1′]. Later constructions control sparsity quantitatively: Kostochka and Rödl obtain high-girth uniform hypergraphs with controlled maximum degree [24, Theorem 3], while Alon, Kostochka, Reiniger, West and Zhu give an explicit construction [25, Theorem 3.2]. Uncountable chromatic number is essential here. For finite hosts, Wang et al. [26, Proposition 1.3(ii)] record

that the weak chromatic numbers of F-free uniform hypergraphs are bounded exactly when F contains no Berge cycle. This does not classify obligatory systems: the expansion C_4^+ contains a Berge cycle and can be avoided by finite triple systems of arbitrarily large chromatic number, yet it occurs in every uncountably chromatic triple system by Theorem A. Finite high-chromatic avoidance and uncountable obligatoriness are therefore different questions.

The proof separates into positive and negative directions. For the positive direction, a probabilistic rainbow lemma provides the local injectivity needed to force complete bipartite expansions. A rooted-abundance lemma then establishes closure under one-point amalgamation, including at singular uncountable cardinals. After the bridges of the Levi graph are deleted, each remaining active component is the expansion of a finite bipartite graph. The quotient is a forest, whose running-intersection property determines a valid amalgamation order.

For the negative direction, we use an independently developed one-apex sequence lift; Li [3, Sections 3–4] gives a parallel formulation. The lift transfers uncountable chromatic number from a graph to a triple system while controlling every finite linear trace. The trace theorem is proved by decomposing the trace into expansion fibres whose support-incidence graph is a forest, so distinct fibres can meet only at cut points. Consequently, neither a bridge-free hyperedge-node nor a Berge cycle can be assembled across fibres. Each of the three failures of the intrinsic criterion has a corresponding host: an Erdős–Rado linear host excludes nonlinearity, the lift of K_{ω_1} excludes a missing bridge, and a classical Erdős–Hajnal graph of uncountable chromatic number whose odd girth exceeds a prescribed finite bound excludes an odd Berge cycle.

All embeddings are injective and non-induced, and the argument is carried out in ZFC. For the classification proof, the imported black-box inputs are the de Bruijn–Erdős compactness theorem [27, Theorem 1, pp. 371–373], the Erdős–Hajnal high-odd-girth theorem [15, Theorem 7.4, p. 76], and the pair case of the Erdős–Rado partition theorem [28, Theorem 4(i), formula (95), p. 471]. The later block-formulation corollary also uses Komjáth's block reduction [18]. For general background on cardinal partition relations, see Erdős et al. [29]; the specific imported statements are those listed above.

1. Preliminaries

A *triple system* is a simple 3-uniform hypergraph $H = (V(H), E(H))$, so $E(H) \subseteq [V(H)]^3$. A colouring $c : V(H) \to \lambda$ is *proper* when no hyperedge is monochromatic, and $\chi(H)$ is the least cardinal λ admitting such a colouring. An embedding of a triple system F in H is an injective map $f : V(F) \to V(H)$ such that $f[e] \in E(H)$ for every $e \in E(F)$; embeddings are not required to be induced. A finite triple system F is *obligatory* if every triple system H with $\chi(H) > \aleph_0$ contains an embedding of F. A point is *isolated* if it belongs to no hyperedge, and F° denotes the result of deleting all isolated points.

All ordinary graphs in this paper are simple. A triple system is *linear* when any two distinct hyperedges meet in at most one point. Its *Levi graph* $I(F)$ is the bipartite graph with classes $V(F)$ and $E(F)$, where a point-node p is adjacent to a hyperedge-node e exactly when $p \in e$. A *bridge* is an edge of a graph whose deletion increases the number of connected components. A *Berge cycle of length* $\ell \geq 2$ consists of distinct points $p_0, \ldots, p_{\ell-1}$ and distinct hyperedges $e_0, \ldots, e_{\ell-1}$ such that $p_i \in e_i \cap e_{i+1}$, with indices modulo ℓ. Equivalently, it is a simple cycle of length 2ℓ in $I(F)$. For graph terminology and the incidence-graph interpretation of hypergraph connectivity, see Diestel [5, Chapters 1 and 3] and Bahmanian and Šajna [30].

For a finite simple graph J, its *private-vertex expansion* J^+ has vertex set

$$V(J) \dot\cup \{p_a : a \in E(J)\}$$

and hyperedges $\{u, v, p_{\{u,v\}}\}$ for $\{u, v\} \in E(J)$, where the points p_a are new and pairwise distinct. A *one-point amalgamation* of vertex-disjoint triple systems F_0, F_1 is obtained by choosing $x_i \in V(F_i)$, identifying x_0 with x_1, and making no other identifications or new hyperedges. Finite disjoint unions and finite edgeless systems include the empty cases.
Finite reductions.

Lemma 1.1 (Isolated-vertex reduction)**.** *Let F be a finite triple system and let F° be obtained by deleting all isolated vertices. Then*

$$F \text{ is obligatory} \iff F^\circ \text{ is obligatory},$$

and

$$F \in \mathcal{B} \iff F^\circ \in \mathcal{B}.$$

Proof. If F° is obligatory and H has uncountable chromatic number, then H has infinitely many vertices. After embedding the finite system F°, map the isolated vertices of F injectively to unused host vertices. Containment is non-induced, so additional host edges are irrelevant. The converse follows because F° is a subhypergraph of F.

For membership in $\mathcal{B}$, the implication $F^\circ \in \mathcal{B} \Rightarrow F \in \mathcal{B}$ follows by taking the disjoint union with a finite edgeless system.

For the converse, use structural induction on a construction of F in $\mathcal{B}$. If $F = J^+$, delete the isolated graph vertices of J to obtain J_0; then $(J^+)^\circ = J_0^+ \in \mathcal{B}$. If F is edgeless, its reduction is the empty edgeless system. Reduction commutes with disjoint union.

Suppose finally that F is a one-point amalgamation of F_0 and F_1, identifying $x_i \in V(F_i)$. By induction, $F_0^\circ, F_1^\circ \in \mathcal{B}$. If both selected vertices are nonisolated in their factors, then F° is the one-point amalgamation of the two reductions at those vertices. If exactly one selected vertex is nonisolated, the isolated side disappears at the identified point and F° is the disjoint union of the two reductions. The same description applies when both selected vertices are

isolated, because the identified point is then deleted. Hence $F^\circ \in \mathcal{B}$ in every case. □

We may therefore suppress isolated vertices in the substantive argument and restore them at the end.

Lemma 1.2 (Finite deletion)**.** *If $\chi(H) > \aleph_0$ and $S \subseteq V(H)$ is finite, then*

$$\chi(H - S) > \aleph_0.$$

Proof. Otherwise a countable colouring of $H - S$, together with finitely many fresh colours for S, would colour H countably. □

Colouring and closure tools.

Lemma 1.3 (Two elementary colouring facts)**.** *We use the following two facts repeatedly.*

1. *If $E(H) = E_1 \cup \cdots \cup E_r$ and every $(V(H), E_i)$ is countably chromatic, then H is countably chromatic.*
2. *If a graph G admits an orientation in which every vertex has outdegree at most $d < \omega$, then $\chi(G) \leq 2d + 1$.*

Proof. For the first assertion, take the product of the r countable colourings.

For the second, every finite subgraph has at most $d|V|$ edges and therefore a vertex of degree at most $2d$. It is $(2d + 1)$-colourable by greedy deletion. The de Bruijn–Erdős compactness theorem [27, Theorem 1] then gives a $(2d + 1)$-colouring of all of G. □

Lemma 1.4 (Closure-chain lemma)**.** *Let κ be an uncountable cardinal, let $1 \leq r < \omega$, and let*

$$\Phi : [\kappa]^r \longrightarrow [\kappa]^{<\omega}$$

have uniformly finite values. There is an increasing continuous chain

$$\varnothing = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_i \subseteq \cdots \qquad (i < \operatorname{cf} \kappa)$$

such that

$$\bigcup_{i<\operatorname{cf}\kappa} M_i = \kappa, \qquad |M_i| < \kappa,$$

and every M_i is Φ-closed:

$$a \in [M_i]^r \quad \Longrightarrow \quad \Phi(a) \subseteq M_i.$$

Proof. Choose sets $A_i \subseteq \kappa$, each of cardinality below κ, whose union is κ. At a successor stage close $M_i \cup A_i$ under Φ, iterating the operation ω times. Since Φ is finitary and finite-valued, this does not increase an infinite cardinal below κ. At a limit stage take the union. A union of an increasing chain of sets closed under a finitary operation remains closed; and because the stage is below $\operatorname{cf} \kappa$, its cardinality remains below κ. □

Write

$$I_i = M_{i+1} \setminus M_i.$$

Then $(I_i)_{i<\operatorname{cf}\kappa}$ partitions κ, and $M_i = \bigcup_{j<i} I_j$.

2. A bipartite subgraph lemma

The following finite complete-bipartite lemma is a consequence of Erdős and Hajnal [15, Corollary 5.6, p. 72]. We give a short closure-chain proof because the construction will also be used later.

Lemma 2.1 (Uncountable chromatic number forces complete bipartite graphs)**.** *For every positive integer n, every graph of uncountable chromatic number contains $K_{n,n}$.*

Proof. Suppose otherwise, and choose a $K_{n,n}$-free graph G of uncountable chromatic number whose vertex cardinality κ is minimal.

For $A \in [\kappa]^n$, let

$$N(A) = \{v \in \kappa : v \text{ is adjacent to every member of } A\}.$$

Because G is $K_{n,n}$-free, $|N(A)| < n$. Apply Lemma 1.4 to $A \mapsto N(A)$, and let M_i, I_i be the resulting chain and layers.

By minimality of κ, every induced graph $G[I_i]$ is countably chromatic. Let $G^\times$ consist of the edges joining distinct layers, and orient each such edge from the later endpoint toward the earlier endpoint. If $v \in I_i$ had n earlier neighbours, those neighbours would form an n-set $A \subseteq M_i$, and closure would give

$$v \in N(A) \subseteq M_i,$$

contrary to $v \in I_i$. Thus the orientation of $G^\times$ has outdegree at most $n-1$. By Lemma 1.3, $G^\times$ is finitely colourable.

Colour each $G[I_i]$ with colours in ω, reusing the same palette on all layers, and combine this with a finite proper colouring of $G^\times$. The resulting countable product colouring is proper on all of G, a contradiction. □

3. Bipartite expansions

Rainbow bipartite submatrices.

Lemma 3.1 (Rainbow bipartite submatrices)**.** *For all positive integers n, t, there exists $q = q(n,t)$ with the following property. Suppose the edges of $K_{q,q}$ are coloured, with an arbitrary colour set, so that at every vertex each individual colour occurs on fewer than t incident edges. Then there are n-element subsets X' and Y' of the two vertex classes such that all n^2 edges between X' and Y' have distinct colours.*

Proof. Choose independent uniformly random n-subsets X' and Y'. For every unordered pair P of distinct same-coloured edges, let I_P be the indicator that both edges of P lie in the selected $K_{n,n}$, and put

$$Z = \sum_P I_P.$$

Thus $Z = 0$ exactly when all selected edges have distinct colours.

Let m_γ be the number of edges of colour γ. The local bound gives

$$m_\gamma \le (t-1)q.$$

At a fixed vertex and for a fixed colour γ, fewer than t incident edges have colour γ. Summing $\binom{d_{v,\gamma}}{2} \le (d_{v,\gamma})(t-1)/2$ over the $2q$ vertices shows that the number of same-coloured pairs sharing a vertex is at most $(t-1)q^2$. The total number of same-coloured pairs is at most

$$\sum_\gamma \binom{m_\gamma}{2} \le \frac{1}{2}(\max_\gamma m_\gamma) \sum_\gamma m_\gamma \le \frac{1}{2}(t-1)q^3.$$

If the two edges share a vertex in the left class, their exact selection probability is

$$\frac{n}{q}\frac{(n)_2}{(q)_2};$$

the same formula holds when the shared vertex is on the right. If the two edges are disjoint, the exact probability is

$$\left(\frac{(n)_2}{(q)_2}\right)^2.$$

For $q \ge 2n$, these are at most $2n^3/q^3$ and $4n^4/q^4$, respectively. By linearity of expectation,

$$\mathbb{E}[Z] \le \frac{2(t-1)n^3}{q} + \frac{2(t-1)n^4}{q}.$$

This is below 1 for sufficiently large q. Since Z is integer-valued and nonnegative, it cannot satisfy $Z \ge 1$ for every choice when $\mathbb{E}[Z] < 1$. Hence some choice has $Z = 0$, and the selected $K_{n,n}$ is rainbow. □

The complete bipartite expansion atom.

Proposition 3.2 (The complete bipartite expansion atom)**.** *For every positive integer n, the triple system $K^+_{n,n}$ is obligatory.*

Proof. Suppose not. Choose a $K^+_{n,n}$-free triple system

$$H = (\kappa, E)$$

with $\chi(H) > \aleph_0$ and with κ minimal.

For a pair xy, write

$$d_H(xy) = |\{z : \{x, y, z\} \in E\}|.$$

Set $t = 3n^2 + 1$ and define a graph R on κ by

$$xy \in E(R) \quad \Longleftrightarrow \quad d_H(xy) \geq t.$$

Claim 3.2.1 (the graph R is countably chromatic).

Otherwise Lemma 2.1 gives a $K_{n,n}$ in R, with core vertex set C, $|C| = 2n$. For each of its n^2 graph edges xy, choose a third vertex p_{xy} with

$$\{x, y, p_{xy}\} \in E,$$

avoiding C and all previously chosen private vertices. At every stage fewer than $2n + n^2 < t$ vertices are forbidden, while xy has at least t possible third vertices. Thus the p_{xy} can be chosen distinct and outside C, producing a copy of $K^+_{n,n}$, a contradiction. □

Let

$$E_1 = \{e \in E : \text{some pair contained in } e \text{ is an edge of } R\}, \qquad E_2 = E \setminus E_1.$$

A proper colouring of R is a proper colouring of (κ, E_1), since every E_1-edge contains an R-edge. Hence $\chi(\kappa, E_1) \leq \aleph_0$, and therefore

$$\chi(\kappa, E_2) > \aleph_0. \tag{3.1}$$

For a pair xy, define

$$\Phi(xy) = \begin{cases} \{z : \{x, y, z\} \in E\}, & d_H(xy) < t, \\ \varnothing, & d_H(xy) \geq t. \end{cases}$$

Apply Lemma 1.4 to Φ, obtaining M_i, I_i.

Every $e \in E_2$ has at least two vertices in its highest layer. Indeed, suppose the highest layer met e only in $z \in I_i$. Then the other pair $xy = e \setminus \{z\}$ lies in M_i. Since $e \in E_2$, $xy \notin R$, so $d_H(xy) < t$. Closure gives $z \in \Phi(xy) \subseteq M_i$, a contradiction.

Thus every E_2-edge is either contained in one layer, or has exactly two vertices in its highest layer I_i and its third vertex in M_i. Let E^*_i be the E_2-edges contained in I_i. Since $|I_i| < \kappa$ and H is $K^+_{n,n}$-free, minimality of κ gives $\chi(I_i, E^*_i) \leq \aleph_0$. Consequently

$$E^* = \bigcup_i E^*_i$$

is countably chromatic. By (3.1), the remaining crossing-edge system

$$E^{**} = E_2 \setminus E^*$$

has uncountable chromatic number.

For each i, define a graph G_i on I_i by declaring $xy \in E(G_i)$ when there is a $\beta \in M_i$ such that $\{x, y, \beta\} \in E^{**}$. If every G_i were countably chromatic, colour each I_i properly in G_i, using the same countable palette on every layer. Every edge of E^{**} would then have differently coloured top-layer vertices, giving a countable colouring of (κ, E^{**}), a contradiction. Hence some G_i is uncountably chromatic.

Choose $q = q(n, t)$ from Lemma 3.1. Lemma 2.1 gives a $K_{q,q}$ in G_i, with sides X, Y. For every $xy \in X \times Y$, choose

$$\beta_{xy} \in M_i$$

such that $\{x, y, \beta_{xy}\} \in E^{**}$, and regard β_{xy} as the colour of the edge xy.

This edge-colouring is locally $(t-1)$-bounded. For example, if for fixed x and fixed β there were t different vertices y with $\beta_{xy} = \beta$, then the pair $\{x, \beta\}$ would have codegree at least t. It would belong to R, forcing the triples $\{x, y, \beta\}$ into E_1, contrary to their membership in $E^{**} \subseteq E_2$. The same argument applies at vertices of Y.

Lemma 3.1 gives n-sets $X' \subseteq X$, $Y' \subseteq Y$ for which the β_{xy}, $x \in X'$, $y \in Y'$, are all distinct. They lie in M_i, whereas $X' \cup Y' \subseteq I_i$, so none is a core vertex. The triples

$$\{x, y, \beta_{xy}\}, \qquad x \in X',\ y \in Y',$$

form $K^+_{n,n}$, the final contradiction. □

Corollary 3.3 (Every finite bipartite expansion is obligatory)**.** *If J is any finite bipartite graph, then J^+ is obligatory.*

Proof. Embed the two vertex classes of J in the two classes of some $K_{n,n}$. Then J^+ is a subhypergraph of $K^+_{n,n}$. Obligatoriness is downward closed under non-induced containment: a copy of the larger finite system contains a copy of the smaller one. Apply Proposition 3.2. □

Corollary 3.3 also follows from Reiher [21, Theorem 1.2]. The argument above gives a direct proof in the notation used here.

4. Closure properties

Closure under disjoint unions and one-point amalgamations is known; see Komjáth [18, pp. 233–238] and the summary in Reiher [21, p. 1]. We include complete proofs because the subsequent argument uses the rooted-abundance construction explicitly.

Disjoint-union closure.

Lemma 4.1 (Disjoint-union closure)**.** *The class of finite obligatory triple systems is closed under finite disjoint unions.*

Proof. Let F_1, F_2 be obligatory and let H be uncountably chromatic. First find a copy of F_1. Delete its finite vertex set. By Lemma 1.2 the remaining triple

system is still uncountably chromatic, so it contains F_2. The two copies are disjoint. Iteration proves the finite case. □

Rooted abundance and one-point amalgamation.

Definition 4.2 (Rooted abundance)**.** Let (F, r) be a finite rooted triple system with $|V(F)| > 1$. For a triple system H and $m \geq 1$, let $R_m(F, r; H)$ be the set of vertices v for which there are m rooted copies of F, all sending r to v, whose off-root vertex sets are pairwise disjoint.

Lemma 4.3 (Rooted-abundance lemma)**.** *If (F, r) is finite, $|V(F)| > 1$, and F is obligatory, and if $\chi(H) > \aleph_0$, then*

$$\chi\big(H[R_m(F, r; H)]\big) > \aleph_0.$$

Proof. Put

$$B = V(H) \setminus R_m(F, r; H).$$

For each $v \in B$, consider the family of off-root vertex sets of all copies of F rooted at v. Because $v \notin R_m(F, r; H)$, this family has matching number at most $m - 1$. Choose a maximal pairwise disjoint subfamily. It contains at most $m - 1$ members; let S_v be their union. Then

$$|S_v| \leq (m - 1)(|V(F)| - 1) =: D.$$

By maximality, S_v meets the off-root part of every rooted copy at v. Otherwise that off-root set would be disjoint from every member of the chosen subfamily and could be added to it.

Form a graph D_B on B by joining v to each member of $S_v \cap B$. If an edge is generated in both directions, assign it to one of its generators, and orient it away from that generator. Every vertex has outdegree at most D, so Lemma 1.3 gives a finite proper colouring of D_B.

Let A be one colour class. Then

$$S_v \cap A = \varnothing \qquad (v \in A).$$

Consequently $H[A]$ contains no copy of F. Indeed, if φ were such a copy and $v = \varphi(r)$, its off-root part would have to meet S_v. All those off-root vertices lie in A, contradicting $S_v \cap A = \varnothing$. Since F is obligatory, every F-free triple system is countably chromatic. Thus each of the finitely many colour classes of D_B induces a countably chromatic subsystem. Tagging a vertex by its finite D_B-colour and its countable colour inside that class gives a countable proper colouring of $H[B]$.

If $H[R_m(F, r; H)]$ were countably chromatic as well, tagging the two sides would give a countable colouring of H. This contradicts $\chi(H) > \aleph_0$. □

Proposition 4.4 (One-point amalgamation closure)**.** *The class of finite obligatory triple systems is closed under one-point amalgamations.*

Proof. Let F_1, F_2 be obligatory, with selected vertices r_1, r_2, and let F be their one-point amalgamation. If one factor consists only of its selected vertex, the assertion is immediate. Otherwise set

$$m = |V(F_2)|.$$

By Lemma 4.3, the set

$$R = R_m(F_1, r_1; H)$$

induces an uncountably chromatic subsystem of every uncountably chromatic H. Hence $H[R]$ contains a copy of F_2; write φ for its embedding and let $v = \varphi(r_2)$.

At v there are m rooted copies of F_1 with pairwise disjoint off-root sets. The set $\varphi(V(F_2) \setminus \{r_2\})$ has $m-1$ vertices, so it meets at most $m-1$ of those off-root sets. One rooted F_1-copy therefore avoids the F_2-copy outside v. Their union is the required one-point amalgamation. □

Corollary 3.3, Lemma 4.1, Proposition 4.4, and the trivial edgeless case prove

$$(2) \Longrightarrow (1). \tag{4.1}$$

5. The intrinsic decomposition

We state the selected-incidence decomposition and its running-intersection argument in the explicit form used by the Lean development. For a distinct structural formulation, compare Li [3, Sections 4–5].

We prove (2) $\Longleftrightarrow$ (3). Isolated vertices may be ignored by Lemma 1.1. Forward preservation.

Proposition 5.1 (The intrinsic conditions are preserved by the generators and operations)**.** *Every generator of $\mathcal{B}$ satisfies the intrinsic conditions, and the conditions are preserved under finite disjoint unions and one-point amalgamations.*

Proof. For a bipartite graph J, the expansion J^+ is linear. Every hyperedge-node is joined to its private point by a bridge. A Berge cycle cannot use a private point, so the Berge cycles of J^+ are precisely the ordinary graph cycles of J, with the same lengths; these are even. Edgeless systems satisfy the conditions vacuously, and disjoint union plainly preserves them.

Consider a one-point amalgamation. Hyperedges belonging to different factors intersect only in the identified point, so linearity is preserved. Its Levi graph is the vertex-sum of the two Levi graphs at a point-node. Every simple cycle in a vertex-sum lies wholly in one factor: a cycle crossing into the other factor would have to pass through the cut vertex twice. Consequently bridges in the factors remain bridges, and every Berge cycle lies in one factor and retains its parity. This uses the standard cut-vertex structure of graph blocks [5,

Chapter 3]; the corresponding hypergraph separation terminology is discussed in Bahmanian and Šajna [30]. □

Thus

$$(2) \Longrightarrow (3). \tag{5.1}$$

Bridge-block reconstruction.

Proposition 5.2 (Bridge-block decomposition)**.** *Suppose F is finite, has no isolated vertices, and satisfies the three intrinsic conditions. Then $F \in \mathcal{B}$.*

Proof. If $E(F) = \varnothing$, then the assumption that F has no isolated vertices gives $V(F) = \varnothing$, and F is an edgeless generator. Hence assume $E(F) \neq \varnothing$. Put $L = I(F)$, let $B(L)$ be its set of bridges, and let $\mathscr{C}$ be the set of connected components of $L - B(L)$.

Consider a hyperedge-node e. Since e is incident with a bridge and has degree three, it has at most two nonbridge incidences. It cannot have exactly one: a nonbridge incidence lies on a cycle, and that cycle must leave e through a second nonbridge incidence. Therefore every hyperedge-node has either no nonbridge incidences or exactly two.
The nontrivial components. Let $C \in \mathscr{C}$ contain a nonbridge incidence. Every hyperedge-node $e \in C$ has exactly two point-neighbours in C. Contract each such e to an ordinary graph edge between those two point-neighbours. This produces a finite graph J_C.

The graph J_C is simple: parallel edges would correspond to two hyperedges of F sharing two points, contrary to linearity. Every cycle in J_C gives a Berge cycle of the same length in F. Hence J_C has no odd cycle and is bipartite.

Each hyperedge-node $e \in C$ has a third point-neighbour p_e across a bridge. These third points satisfy:

1. $p_e \notin C$; otherwise a path in C from p_e to e, together with the incidence ep_e, would place that incidence on a cycle.
2. If $e \neq f$ lie in C, then $p_e \neq p_f$; otherwise a path in C from e to f, together with ep_e and $p_f f$, would place those bridge incidences on a cycle.

Define a map from J_C^+ to the subsystem P_C formed by these hyperedges as follows. Send every graph vertex of J_C to the corresponding point-node of C, send the private vertex indexed by the contracted edge associated with e to p_e, and send that expanded edge to the hyperedge e of F. The two observations above show that the vertex map is injective: the private points are pairwise distinct and lie outside C. Every e has exactly the two contracted endpoints and p_e as its three vertices, so this map is incidence-preserving and surjective on the vertices and hyperedges of P_C. Hence it is an isomorphism $J_C^+ \cong P_C$.
Components consisting of one hyperedge-node. If a hyperedge-node e has no nonbridge incidence, it is an isolated component after the bridges are deleted. Its single triple is the expansion of a one-edge bipartite graph. Thus every hyperedge of F belongs to a piece P_C isomorphic to J^+ for a finite bipartite

graph J. The hyperedge sets of these pieces partition $E(F)$. Their vertex sets cover $V(F)$: every point is incident with a hyperedge, and it either lies in that hyperedge-node's bridge-deleted component or is the point endpoint of a bridge leaving it.
Quotient forest. Contract every component of $L - B(L)$. The original bridges become edges of a quotient multigraph T. Two distinct bridges cannot join the same pair of bridge-deleted components: paths inside those two components, together with the two bridges, would form a cycle containing both, contradicting that they are bridges. Thus T is a simple graph.

To see that T is a forest, suppose that it contained a cycle. Within each contracted component on this cycle, choose a path joining the endpoints of the two incident quotient edges. Concatenating these paths with the corresponding bridges gives a closed walk in L that traverses each of those bridges exactly once. This is impossible: deleting a bridge separates its endpoints, so every closed walk crosses the resulting cut an even number of times. Hence T is a forest.

A vertex $C \in V(T) = \mathscr{C}$ is called *active* if the bridge-deleted component C contains a hyperedge-node. For an active C, the piece P_C consists of the hyperedges represented in C, all point-nodes of C, and the point endpoints of bridge incidences leaving hyperedge-nodes in C. An inactive component contains no hyperedge-node and hence no incidence of $L - B(L)$, so it is a singleton point-node.

Every hyperedge now belongs to a bipartite expansion piece. It remains to order the pieces so that each new piece meets the assembled system in at most one point. The quotient forest records the attachment geometry needed for this running-intersection property.

Claim (5.2.1. Running intersection)**.** *Every component of T contains an active vertex, because F has no isolated vertices. Choose an active root in each component, and order the active vertices by nondecreasing depth, with arbitrary order among equal-depth vertices. When an active component C is added, its piece P_C meets the union of the earlier pieces in at most one point.*

Proof of the claim. For a point $p \in V(F)$, let $X(p) \in \mathscr{C}$ be the component of $L - B(L)$ containing the point-node p. The active pieces containing p are precisely $P_{X(p)}$, when $X(p)$ is active, together with the pieces P_D at active neighbours D of $X(p)$ whose corresponding original bridge has point endpoint p. Thus the vertices of T indexing pieces that contain p are contained in the closed star centred at $X(p)$.

Suppose that $p \in P_C$ also lies in an earlier piece. If $X(p) = C$, every other piece containing p is indexed by a neighbour of C. A tree has only one neighbour of C at smaller depth, namely its parent, so only that piece can be earlier. If $X(p) \neq C$, then C is a neighbour of $X(p)$. The vertex $X(p)$ cannot be a child of C, because then both $X(p)$ and every other vertex in its closed star would be deeper than C. Hence $X(p)$ is the parent of C. In either case, p is the point

endpoint of the unique parent edge of C. The same reasoning applies to every point shared with an earlier piece, so all such points coincide with this parent point p_C, and

$$P_C \cap \bigcup_{D \text{ earlier}} P_D \subseteq \{p_C\}.$$

For a root active component the intersection is empty. □

Starting with the root piece in each component, add the active pieces in this order. If the new intersection is empty, use a disjoint union. If it is the singleton $\{p_C\}$, use a one-point amalgamation at that point. Several earlier pieces may already contain the same inactive attachment point, but the new step still identifies only that one existing vertex.

The identity maps from the pieces into F assemble inductively to an incidence isomorphism onto the union of the pieces already added. By the running-intersection claim, the next piece and the preceding union are either disjoint or meet exactly at the point used in the amalgamation, so no unintended identification is introduced. The piece hyperedge sets are disjoint and partition $E(F)$, and their vertex sets cover $V(F)$. After all active pieces have been added, the resulting subsystem is F. Different components of T are combined by disjoint union, proving $F \in \mathcal{B}$. □

This proves

$$(3) \Longrightarrow (2). \tag{5.2}$$

Combining (5.1) and (5.2),

$$(2) \Longleftrightarrow (3). \tag{5.3}$$

Theorem 5.3 (Canonical atom normal form)**.** *Let F be a finite triple system without isolated points satisfying the three intrinsic conditions in Theorem A. Its hyperedges admit a canonical partition into* atoms *with the following properties.*

(1) Every atom is either one triple or J^+ for a finite 2-connected simple bipartite graph J.
(2) Two distinct atoms meet in at most one point. The bipartite incidence graph between atoms and points belonging to at least two atoms is a forest.
(3) The atom partition, the nontrivial cores J up to graph isomorphism, and the atom–point forest are determined by F.

Conversely, every forest assembly of these atoms by disjoint unions and one-point amalgamations satisfies the intrinsic conditions.

Proof. Let $L = I(F)$. A *cyclic block* is a maximal 2-connected subgraph of L containing a cycle. Since every hyperedge-node has degree three and is incident with a bridge, its degree inside a cyclic block is two. Suppressing the hyperedge-nodes in such a block C gives a finite graph J_C on the point-nodes of C. It is

simple by linearity and 2-connected by standard block theory [4, 5]. Its cycles correspond to Berge cycles of the same lengths, so J_C is bipartite.

For a hyperedge-node $e \in C$, the third incidence is a bridge. Its point endpoint lies outside C, since otherwise that incidence would lie on a cycle. These third points are pairwise distinct as e varies in C; otherwise a path inside C between two hyperedge-nodes, completed through the common third point, would put both bridge incidences on a cycle. The hyperedges represented in C therefore form exactly J_C^+.

A hyperedge-node outside every cyclic block has no nonbridge incidence and contributes one single-triple atom. Thus the atoms partition the hyperedges. Two atoms cannot share two points, since paths inside the two atoms would produce a Levi cycle crossing two cyclic blocks. Similarly, a cycle in the atom–shared-point incidence graph expands to a closed Levi walk and hence to a Levi cycle crossing atom boundaries. The incidence graph is therefore a forest. Cyclic blocks and the block–cut forest are canonical, proving the forward statement. The converse follows because an atomic expansion is linear, has a private bridge at each hyperedge-node, and has only even Berge cycles, while a forest of one-point sums preserves these properties. □

The graph primitives in this normal form are classical algorithmic objects: bridges and bridge-connected components can be found by Tarjan's algorithm [31], while biconnected components admit a depth-first-search construction [32]. These algorithms are distinct from the structural identification above; no verified running-time bound for the Lean development is asserted.

Call a connected triple system with at least one hyperedge *one-point indecomposable* if it cannot be written as a one-point amalgamation of two systems that both contain a hyperedge.

Corollary 5.4 (Minimal generators and one-point indecomposables)**.** *Up to isomorphism, the class $\mathcal{B}$ is generated by finite edgeless systems, one triple, and J^+ for finite 2-connected simple bipartite graphs J, under finite disjoint unions and one-point amalgamations. A connected reduced obligatory system with at least one hyperedge is one-point indecomposable if and only if it is one triple or J^+ for a finite 2-connected simple bipartite graph J.*

Proof. The theorem decomposes every reduced member of $\mathcal{B}$ into the displayed atoms; isolated points are restored by an edgeless factor. If the atom forest has at least two atom-nodes, a leaf atom separates from the rest at its unique shared point. Conversely, one triple is indecomposable, and if J is 2-connected then deleting any point of J^+ cannot separate its hyperedge-nodes into two nonempty classes: a private point is a leaf, while a core point leaves J connected. Hence J^+ is indecomposable. □

6. The sequence lift

We use a one-apex construction specialised to ω_1, presented through base fibres and their support-incidence graph to match the formalisation. Li [3,

Sections 3–4 and Theorem 4.6] gives an alternative one-apex and bridge-trace formulation.

The one-apex construction. Let $G = (X, A)$ be an arbitrary graph, where $A = E(G)$ is also regarded as an alphabet. Let

$$T = A^{<\omega_1}$$

be the tree of all sequences of graph edges of countable ordinal length. For $s \in T$ and $x \in X$, write $x_s = (s, x)$.

Definition 6.1 (The one-apex lift)**.** Define the triple system $\mathcal{L}(G)$ by

$$V(\mathcal{L}(G)) = T \times X.$$

For $s \in T$, an edge $a = \{x, y\} \in A$, a sequence $t \in T$ extending $s^\frown a$, and $z \in X$, put

$$\{x_s, y_s, z_t\} \in E(\mathcal{L}(G)). \tag{6.1}$$

The two vertices x_s, y_s form the base pair and z_t is the apex. Since t properly extends s, these are three distinct vertices. In the resulting triple, the sequence node occurring twice uniquely identifies s; its two graph labels identify the edge $a = \{x, y\}$, and the remaining vertex identifies t, z. Thus each hyperedge arises from unique data, up to interchanging x and y, and the system is simple and 3-uniform.

Chromatic and finite-trace properties.

Lemma 6.2 (Chromatic lower bound)**.** *If $\chi(G) > \aleph_0$, then*

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

Proof. Suppose $c : T \times X \to \omega$ is proper. For every $s \in T$, the restriction $x \mapsto c(x_s)$ is a countable colouring of G. Hence there is an edge

$$a_s = \{x_s^0, x_s^1\} \in E(G)$$

whose two vertices at node s have a common colour k_s.

Construct $s_\alpha \in A^\alpha$ for $\alpha < \omega_1$ by

$$s_0 = \varnothing, \qquad s_{\alpha+1} = s_\alpha^\frown a_{s_\alpha},$$

and at limit ordinals take unions.

If $\alpha < \beta$, then s_β extends $s_\alpha^\frown a_{s_\alpha}$. If $k_{s_\alpha} = k_{s_\beta}$, take either endpoint z of the monochromatic edge at s_β. By (6.1),

$$\{(s_\alpha, x_{s_\alpha}^0), (s_\alpha, x_{s_\alpha}^1), (s_\beta, z)\}$$

is a monochromatic hyperedge. Therefore the colours k_{s_α}, $\alpha < \omega_1$, are pairwise distinct, impossible in ω. □

The following trace theorem uses the fibre decomposition described above. Li gives a parallel bridge-trace formulation [3, Theorem 4.6].

Theorem 6.3 (Finite linear trace decomposition)**.** *Every finite, not necessarily induced, linear subhypergraph K of $\mathcal{L}(G)$ is obtainable, by finite disjoint unions and one-point amalgamations, from systems J^+ where J is a finite subgraph of G.*

Proof. We use three local objects in the assembly. A *base node* is a sequence s at which a lift hyperedge begins. The two vertices with sequence coordinate s are its *core vertices*; the third vertex, whose coordinate properly extends s, is its *apex*. The *base fibre* K_s consists of all hyperedges of K based at s, together with their incident vertices. A base node is *active* when its fibre is nonempty. We first identify each fibre with an expansion piece and then show that the fibres form a forest of one-point overlaps.

First set aside every isolated vertex of K. Each is a one-vertex edgeless factor, hence isomorphic to J^+ for a one-vertex edgeless subgraph J of G, and may be restored by disjoint union at the end. We may therefore assume that every vertex of K lies in a hyperedge.

Every hyperedge of $\mathcal{L}(G)$ has a unique base node: exactly two of its vertices have sequence coordinate s, while the apex has a strictly longer sequence coordinate. We group the hyperedges of K by this coordinate and use the notation K_s from the dictionary.

Claim 6.3.1 (each base fibre is one expansion piece). For every active base node s, there is a finite subgraph J_s of G such that $K_s \cong J_s^+$.

Proof of Claim 6.3.1. For a fixed graph edge $a = \{x, y\} \in E(G)$, at most one hyperedge of K_s can have base pair $\{x_s, y_s\}$, since two such hyperedges would share two vertices and violate linearity. Let J_s be the finite subgraph of G whose edges are precisely the graph edges used by hyperedges of K_s, with vertex set consisting of their endpoints.

Define a map $J_s^+ \to K_s$ by sending each graph vertex $x \in V(J_s)$ to the core vertex x_s, and by sending the private vertex of $a = \{x, y\}$ to the apex of the unique hyperedge of K_s with base pair $\{x_s, y_s\}$. Distinct graph edges have distinct apices: if z_t were the apex for a and a', the coordinate of t immediately after s would have to equal both a and a'. No apex is a core vertex, because an apex has sequence coordinate strictly extending s, whereas every core vertex has coordinate s. The map is therefore injective, and by construction it is bijective on hyperedges and preserves each incidence. Hence $K_s \cong J_s^+$. □

Claim 6.3.2 (two fibres have at most one common point). For distinct active base nodes s and u,

$$|V(K_s) \cap V(K_u)| \leq 1.$$

Proof of Claim 6.3.2. If a point z_t lies in both K_s and K_u, then both s and u are prefixes of t, so they are comparable. Suppose $s \subsetneq u$. A common point cannot be a core point of K_s; it is an apex of K_s. The graph edge used by its hyperedge in K_s is the coordinate of u immediately following s. Hence all points in $K_s \cap K_u$ would be apices of the same base edge of K_s. There is at most one such apex. Therefore

$$|V(K_s) \cap V(K_u)| \leq 1. \tag{6.2}$$

□

Let Q be the bipartite incidence graph of the base fibres. Its left class is the set of active base nodes. Its right class consists of points that lie in at least two fibres. Join a base node s to a point p exactly when $p \in V(K_s)$.

Claim 6.3.3 (the overlap graph is a forest). The graph Q is acyclic.

Proof of Claim 6.3.3. Suppose that Q has a cycle, and choose on it a base node s of minimum sequence length. Its two neighbouring base nodes on the cycle are proper descendants of s. The two intervening shared points are distinct, so they are distinct apices in K_s, arising from two distinct graph edges $a \neq a'$. Thus the two neighbouring base nodes lie in different immediate branches below s, one beginning with a and the other with a'.

Starting from one neighbour of s, every base node on the remaining part of the cycle lies in the same immediate branch below s. Indeed, consecutive base nodes share a point and are therefore comparable. For two comparable proper descendants of s, the first sequence coordinate after s is the same, so a length-two step through a shared point cannot move from one immediate branch to another. Induction along the remaining path therefore keeps every base node in the first branch, contradicting that its final node lies in the other branch. Hence Q is acyclic. □

Root each component of Q at a base node, and order its base nodes by nondecreasing even distance from the root. For a nonroot base node s, let p_s be the point-node adjacent to s on the path to the root.

Claim (6.3.4. Base-fibre running intersection)**.** *Every earlier factor that meets K_s meets it at p_s.*

Proof of Claim 6.3.4. Suppose K_u is earlier and contains a point $p \in K_s$. Then Q contains the length-two path $s - p - u$. Let the distance of s from the root be $2d$. The point-node p has distance either $2d - 1$ or $2d + 1$. In the second case u has distance $2d + 2$, so it cannot be earlier. Thus p has distance $2d - 1$, and the uniqueness of the path to the root in the tree Q gives $p = p_s$. This also covers an earlier sibling u, which shares the same parent point p_s. □

Claim 6.3.5 (the ordered fibres assemble to K). Let U_s be the union of the factors preceding K_s. Equation (6.2) and Claim 6.3.4 give

$$V(K_s) \cap U_s \subseteq \{p_s\}.$$

Starting with the root factor, add the remaining factors in this order by disjoint unions or one-point amalgamations as appropriate. At each step the natural maps into K identify exactly the displayed common point and nothing else, so they assemble to an isomorphism onto the union of the factors already added. Different components of Q, and the isolated factors removed at the start, are combined by disjoint union. This is the running-intersection step: Claim 6.3.1 supplies the expansion pieces, while Claims 6.3.2–6.3.4 certify that each new piece is attached along at most one previous point. □

Corollary 6.4 (Restrictions on finite linear traces)**.** *Every finite linear $K \subseteq \mathcal{L}(G)$ satisfies:*

1. *every hyperedge-node of $I(K)$ is incident with a bridge;*
2. *every Berge cycle of K is contained in a single factor J_s^+;*
3. *a Berge cycle of length ℓ in K yields an ordinary cycle of length ℓ in G.*

Proof. In every J_s^+ the incidence with the private vertex is a bridge. Bridgehood is preserved by one-point amalgamation. A simple Levi cycle cannot cross a one-point cut, so every Berge cycle lies in one factor. Berge cycles in J_s^+ are exactly ordinary graph cycles in J_s. □

7. A classical high-odd-girth input

For the odd-cycle obstruction we use the following classical graph theorem directly, without passing through a shift-graph surrogate.

Theorem 7.1 (Erdős–Hajnal)**.** *For every positive integer m there is a graph G such that*

$$\chi(G) > \aleph_0$$

and G contains no odd cycle of length at most m.

Source. Theorem 7.4 of Erdős and Hajnal [15, p. 76], as quoted in Erdős et al. [17, Theorem C, p. 428], gives for every positive integer i an uncountably chromatic graph containing no cycle C_{2j+1} for $0 < j < i$. Choose i so that $2i - 1 \geq m$. □

8. Avoidance hosts

The intrinsic criterion can fail in three ways: nonlinearity, a missing bridge, or an odd Berge cycle. Li [3, Sections 2–4] organises the negative direction in a parallel way. We prove the first two avoidance constructions in our notation and apply the Erdős–Hajnal theorem from Section 7 to the third.

We prove

$$\neg(3) \Longrightarrow \neg(1).$$

Nonlinearity obstruction.

Proposition 8.1 (Avoidance of every nonlinear finite triple system)**.** *There is an uncountably chromatic linear triple system. Consequently, every finite nonlinear triple system is non-obligatory.*

Proof. We use the Erdős–Rado relation [28, Theorem 4(i), formula (95), p. 471]

$$(2^{\aleph_0})^+ \longrightarrow (\aleph_1)^2_{\aleph_0}. \tag{8.1}$$

Let $\kappa = (2^{\aleph_0})^+$. Define a triple system T_κ whose vertices are the pairs in $[\kappa]^2$, and whose hyperedges are

$$\{\{\alpha,\beta\},\{\alpha,\gamma\},\{\beta,\gamma\}\}, \qquad \alpha<\beta<\gamma<\kappa.$$

This triple system is linear: two distinct triangles of a complete graph cannot share two graph edges, because two graph edges of a triangle determine the third edge and the three underlying vertices.

It is uncountably chromatic. A countable vertex-colouring of T_κ is a countable edge-colouring of K_κ. By (8.1) there is a monochromatic subset of cardinality $\aleph_1$, and in particular a monochromatic graph triangle. This is a monochromatic hyperedge of T_κ.

If a finite triple system F is nonlinear, it has two distinct edges sharing at least two vertices. No injective embedding of those edges can exist in the linear host T_κ. Thus T_κ is F-free. □

Missing-bridge obstruction.

Proposition 8.2 (Avoidance when the bridge condition fails)**.** *Let F be finite and linear, and suppose some hyperedge-node of $I(F^\circ)$ is incident with no bridge. Then F is non-obligatory.*

Proof. This is the K_{ω_1} specialisation of the sequence lift defined above. Li gives an alternative missing-bridge obstruction [3, Section 4]. By Lemma 1.1 it suffices to avoid F°, so assume that F has no isolated vertices. Take $G = K_{\omega_1}$. Then $\chi(G) > \aleph_0$, and Lemma 6.2 gives

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

If F appeared in $\mathcal{L}(G)$, retain exactly the host hyperedges corresponding to the edges of F. Because the embedding is injective and F is linear, these selected host hyperedges form a finite linear subhypergraph isomorphic to F. Corollary 6.4 says that every hyperedge-node of such a trace is incident with a bridge, contradicting the chosen hyperedge of F. Therefore $\mathcal{L}(K_{\omega_1})$ is an uncountably chromatic F-free host. □

Odd-cycle obstruction.

Proposition 8.3 (Avoidance of an odd Berge cycle)**.** *Let F be finite and linear, and suppose F° has an odd Berge cycle. Then F is non-obligatory.*

Proof. Suppress isolated vertices and put $m = |E(F)|$. By Theorem 7.1 choose a graph G with $\chi(G) > \aleph_0$ and no odd cycle of length at most m. Lemma 6.2 gives $\chi(\mathcal{L}(G)) > \aleph_0$.

Suppose that F embeds in $\mathcal{L}(G)$ and retain exactly the host hyperedges corresponding to the edges of F. The selected trace is finite and linear. Corollary 6.4 localises every Berge cycle in one expansion fibre and sends a Berge cycle of length ℓ to an ordinary cycle of the same length in G. The given odd Berge cycle has $\ell \le m$, contradicting the choice of G. Thus $\mathcal{L}(G)$ is an uncountably chromatic F-free host. □

The three propositions cover every failure of the intrinsic conditions and therefore prove

$$\neg(3) \Longrightarrow \neg(1). \tag{8.2}$$

9. Proof of the classification

Sections 3 and 4 establish (2) $\Longrightarrow$ (1), Section 5 establishes (2) $\Longleftrightarrow$ (3), and Section 8 proves the contrapositive $\neg(3) \Longrightarrow \neg(1)$. Together these implications prove all equivalences in Theorem A. In particular,

$$F \text{ is obligatory} \iff F \in \mathcal{B},$$

$$F \in \mathcal{B} \iff \begin{array}{l} F^\circ \text{ is linear,} \\ \text{every hyperedge-node of } I(F^\circ) \text{ is incident with a bridge,} \\ \text{every Berge cycle of } F^\circ \text{ has even length.} \end{array}$$

All embeddings are non-induced, so additional host hyperedges on the image vertices are irrelevant. The proof uses only ZFC, graph-colouring compactness [27], the classical Erdős–Hajnal high-odd-girth theorem [15], and the standard Erdős–Rado relation (8.1) [28]. It assumes neither CH nor GCH and uses no forcing axiom or large-cardinal hypothesis.

10. Finite parameter consequences

In this section, a triple system is *connected* when its Levi graph is connected. We first exclude isolated vertices. For an integer $r \ge 1$, set

$$q(r) = \lceil 2\sqrt{r} \rceil .$$

Proposition 10.1 (Edge-deletion form of the bridge condition)**.** *Let F be a finite triple system and let $e = \{x, y, z\} \in E(F)$. The hyperedge-node e has no incident bridge in $I(F)$ if and only if x, y, z lie in one connected component after the hyperedge e is deleted and all points are retained. Consequently the*

bridge condition in Theorem A is equivalent to the following hypergraph-native condition:

$$\textit{for every } e \in E(F^\circ),\ \textit{deleting } e$$
$$\textit{separates its three points into at least two components.}$$

Proof. Fix $x \in e$. The Levi incidence xe is not a bridge precisely when there is an alternative path from e to x after that incidence is deleted. A simple such path leaves e through y or z and thereafter lies entirely in the Levi graph of $F - e$. Thus xe is non-bridging exactly when x is connected in $F - e$ to at least one of the other two points of e.

If all three incidences at e are non-bridging, no one of x, y, z can form a singleton component among those three points. A partition of three points with this property has one block, so all three are connected in $F - e$. The converse follows by reversing the same paths. □

Lemma 10.2 (Bipartite shadow)**.** *Let F be an obligatory finite triple system with no isolated vertices and at least one hyperedge. There is a finite bipartite graph J, without isolated vertices, such that*

$$|E(J)| = |E(F)|, \qquad |V(J)| = |V(F)| - |E(F)|,$$

and J has the same number of connected components as F. Conversely, for every finite bipartite graph J without isolated vertices, the expansion J^+ is obligatory and has these parameter relations.

Proof. By Theorem A and Proposition 5.2, each connected component H of F is obtained by one-point amalgamations from connected expansion pieces

$$J_1^+, \dots, J_k^+,$$

where each J_i is a connected bipartite graph. Disconnected graph pieces may simply be split into their connected components. Write $m_i = |E(J_i)|$ and $s_i = |V(J_i)|$. Since H is connected, its k pieces are joined by exactly $k - 1$ one-point identifications. Hence

$$|E(H)| = \sum_{i=1}^{k} m_i, \qquad |V(H)| = \sum_{i=1}^{k} (m_i + s_i) - (k - 1). \tag{10.1}$$

Take one-point sums of the ordinary graphs $J_1, \dots, J_k$ along any tree on the k indices. Before each identification, interchange the two bipartition classes of the new factor if necessary. The resulting graph J_H is connected and bipartite, and (10.1) gives

$$|E(J_H)| = |E(H)|, \qquad |V(J_H)| = |V(H)| - |E(H)|.$$

Taking the disjoint union of these shadows over the components of F proves the forward assertion. The converse is the expansion-generator case of Theorem A, with the displayed counts immediate from the definition of J^+. □

Theorem 10.3 (Exact order–size–component spectrum)**.** *Let $m \geq 1$, $1 \leq c \leq m$, and n be integers. There exists an obligatory triple system F with no isolated vertices, exactly m hyperedges, exactly n vertices, and exactly c connected components if and only if*

$$\boxed{m + 2(c-1) + \left\lceil 2\sqrt{m-c+1} \right\rceil \leq n \leq 2m + c.} \tag{10.2}$$

Every integer in this interval occurs.

Proof. We begin with the corresponding elementary graph fact. A connected simple bipartite graph with $r \geq 1$ edges and s vertices exists if and only if

$$q(r) \leq s \leq r + 1. \tag{10.3}$$

Indeed, connectedness gives $r \geq s - 1$. If the bipartition has sizes a, b, then

$$r \leq ab \leq \left\lfloor \frac{s^2}{4} \right\rfloor,$$

which is equivalent to $s \geq q(r)$. Conversely, $K_{\lfloor s/2 \rfloor, \lceil s/2 \rceil}$ contains a spanning tree, and starting with that tree and adding arbitrary unused edges realises every edge count between $s - 1$ and $\lfloor s^2/4 \rfloor$.

Apply Lemma 10.2. Let the c connected components of the shadow have positive edge counts $m_1, \ldots, m_c$, where $\sum_i m_i = m$. If their orders are s_i, then (10.3) gives

$$\sum_{i=1}^{c} q(m_i) \leq \sum_{i=1}^{c} s_i \leq m + c. \tag{10.4}$$

For positive integers a, b,

$$q(a) + q(b) \geq q(a + b - 1) + 2. \tag{10.5}$$

To see this, note that

$$\sqrt{a} + \sqrt{b} \geq \sqrt{a + b - 1} + 1,$$

because, after squaring, the assertion reduces to $(a-1)(b-1) \geq 0$; taking ceilings gives (10.5). Repeatedly applying (10.5) to the left side of (10.4) yields

$$\sum_{i=1}^{c} q(m_i) \geq q(m - c + 1) + 2(c - 1).$$

The shadow has $n - m$ vertices, so these inequalities give (10.2).

For sharpness put $M = m - c + 1$. Take $c - 1$ components equal to K_2. By (10.3), a final connected bipartite component with M edges can have every order from $q(M)$ through $M + 1$. Their disjoint union therefore realises every shadow order from $q(M) + 2(c-1)$ through $m + c$. Expanding it adds exactly m private vertices and proves attainability of every integer in (10.2). □

Corollary 10.4 (Connected and unrestricted spectra)**.** *For $m \geq 1$:*

(1) *a connected obligatory triple system without isolated vertices has m hyperedges and n vertices if and only if*

$$m + \lceil 2\sqrt{m} \rceil \le n \le 2m + 1;$$

(2) *without prescribing the number of components, a reduced obligatory triple system has m hyperedges and n vertices if and only if*

$$m + \lceil 2\sqrt{m} \rceil \le n \le 3m;$$

(3) *if isolated vertices are permitted, an obligatory triple system with m hyperedges and n vertices exists if and only if*

$$n \ge m + \lceil 2\sqrt{m} \rceil .$$

Proof. The first assertion is Theorem 10.3 with $c = 1$. For the second, the connected interval covers through $2m + 1$; for $n > 2m + 1$, take $c = n - 2m$ and use the upper endpoint $n = 2m + c$ of (10.2). The largest possible value is obtained from m disjoint triples. The final assertion follows by adjoining arbitrary isolated vertices. □

Corollary 10.5 (Size spectrum at fixed order)**.** *Let $c \ge 1$ and $n \ge 3c$. A reduced obligatory triple system with exactly n vertices, c connected components, and m hyperedges exists if and only if*

$$\boxed{\left\lceil \frac{n-c}{2} \right\rceil \le m \le n - 2c + 4 - \left\lceil 2\sqrt{n - 3c + 4} \right\rceil .} \tag{10.6}$$

Consequently, a reduced obligatory triple system with exactly c connected components and n vertices exists if and only if

$$n = 3c, \qquad n = 3c + 2, \qquad \text{or} \qquad n \ge 3c + 4. \tag{10.7}$$

Proof. The upper-order inequality in (10.2) is equivalent to $m \ge \lceil (n - c)/2 \rceil$. For the other inequality put

$$M = m - c + 1, \qquad N = n - 3c + 3.$$

It becomes

$$M + \left\lceil 2\sqrt{M} \right\rceil \le N. \tag{10.8}$$

For integers $N \ge 3$ and $M \ge 1$, the largest M satisfying (10.8) is

$$U = N + 2 - \left\lceil 2\sqrt{N + 1} \right\rceil . \tag{10.9}$$

Indeed, write $k = \lceil 2\sqrt{N + 1} \rceil$. Since $k^2 \ge 4(N + 1)$,

$$4U = 4N + 8 - 4k \le (k - 2)^2,$$

so $\lceil 2\sqrt{U} \rceil \le k - 2$ and $U + \lceil 2\sqrt{U} \rceil \le N$. On the other hand, $(k - 1)^2 < 4(N + 1)$ gives

$$(k - 3)^2 < 4(N + 3 - k) = 4(U + 1),$$

so $\lceil 2\sqrt{U + 1} \rceil \ge k - 2$ and $(U + 1) + \lceil 2\sqrt{U + 1} \rceil \ge N + 1$. The left side of (10.8) is strictly increasing in M, proving (10.9). Translating back to m gives (10.6).

For (10.7), Corollary 10.4 shows that a connected reduced obligatory system has order 3, order 5, or any order at least 7. The first two statements come from $m = 1, 2$. For $m \geq 3$, the connected intervals have no gaps because $\lceil 2\sqrt{m+1} \rceil \leq m+1$ for $m+1 \geq 4$. Thus every component has order

$$3 + r, \qquad r \in \{0, 2\} \cup [4, \infty).$$

The sum of c such increments is again 0, 2, or any integer at least 4: use one nonzero increment and take all remaining increments to be zero. The increments 1 and 3 are impossible. This proves (10.7). □

Corollary 10.6 (Exact Levi cycle-rank spectrum)**.** *Let F be reduced and obligatory, with m hyperedges, n vertices, and c connected components. The cyclomatic number of its Levi graph is*

$$\beta(I(F)) = |E(I(F))| - |V(I(F))| + c = 2m - n + c. \tag{10.10}$$

For fixed m, c, every integer in the interval

$$0 \leq \beta(I(F)) \leq m - c + 2 - \left\lceil 2\sqrt{m-c+1} \right\rceil \tag{10.11}$$

occurs, and no other value occurs. In particular,

$$|V(F)| = 2|E(F)| + c \quad \Longleftrightarrow \quad I(F) \text{ is a forest.}$$

Proof. The Levi graph has $n+m$ vertices, $3m$ incidence edges, and c components, which gives (10.10). Substitution of the two endpoints in (10.2) gives (10.11), and every intermediate value occurs because every intermediate order does. A finite graph has cyclomatic number zero exactly when it is a forest. □

Corollary 10.7 (Rigidity at balanced lower endpoints)**.** *Let F be connected, reduced, and obligatory.*

(1) If $|E(F)| = t^2$ and $|V(F)| = t^2 + 2t$, then $F \cong K^+_{t,t}$.

(2) If $|E(F)| = t(t+1)$ and $|V(F)| = t(t+1) + 2t + 1$, then $F \cong K^+_{t,t+1}$.

Here $t \geq 1$, and the complete bipartite graphs are unique up to interchanging the two classes.

Proof. In either case the shadow from Lemma 10.2 attains $|E(J)| = \lfloor |V(J)|^2/4 \rfloor$. Equality forces every possible cross edge to be present and the two bipartition classes to be balanced, so the shadow is $K_{t,t}$ or $K_{t,t+1}$, respectively.

For $t \geq 2$ these complete bipartite graphs have no cut vertex. On the other hand, the shadow built from two or more connected expansion pieces is a nontrivial one-point sum and has a cut vertex. Hence the bridge-block reconstruction has only one piece, and F is the expansion of the displayed complete bipartite graph. The cases $t = 1$ are immediate. □

Canonical atoms and structural spectra. For a connected reduced obligatory system F, put

$$s = |V(F)| - |E(F)|, \qquad \beta = 2|E(F)| - |V(F)| + 1.$$

Thus s is the order of a connected bipartite shadow and β is its cycle rank. Let $a(F)$ denote the number of canonical atoms from Theorem 5.3.

Lemma 10.8 (Minimum order at fixed cyclic rank)**.** *Let J be a finite 2-connected simple bipartite graph of cycle rank $r = |E(J)| - |V(J)| + 1 \geq 1$. Then*

$$|V(J)| \geq 2 + q(r). \tag{10.12}$$

For $r \geq 2$, every order $v \geq 2 + q(r)$ occurs. For $r = 1$, the possible orders are exactly the even integers $v \geq 4$.

Proof. Writing $v = |V(J)|$, the bipartite extremal bound gives

$$r \leq \left\lfloor \frac{v^2}{4} \right\rfloor - v + 1 = \left\lfloor \frac{(v-2)^2}{4} \right\rfloor,$$

which is equivalent to (10.12). If v is even, start with the spanning cycle C_v in the balanced complete bipartite graph and add edges until the desired rank is reached. If v is odd, start with C_{v-1} and add the remaining vertex with two neighbours in the opposite bipartition class; this is 2-connected and has rank two. Add arbitrary missing cross-edges. These constructions realise every allowed $r \geq 2$. A 2-connected graph of rank one is a cycle, which is bipartite exactly when its order is even. □

Theorem 10.9 (Exact canonical atom-count spectrum)**.** *Fix nonnegative integers s, β satisfying either $\beta = 0$ and $s \geq 2$, or $\beta \geq 1$ and $s \geq 2 + q(\beta)$. Let F range over connected reduced obligatory systems with at least one hyperedge, shadow order s and cycle rank β. The possible values $k = a(F)$ are exactly*

$$\begin{aligned} &\beta = 0: \quad k = s - 1, \\ &\beta = 1: \quad 1 \leq k \leq s - 3, \quad k \equiv s + 1 \pmod 2, \\ &\beta \geq 2: \quad 1 \leq k \leq s - 1 - q(\beta). \end{aligned} \tag{10.13}$$

For $\beta \geq 1$, the maximum is

$$a_{\max}(s, \beta) = s - 1 - q(\beta). \tag{10.14}$$

Every maximiser has one minimum-order cyclic atom carrying the full rank β, and every other atom is one triple. The parity obstruction in the rank-one line is the only gap in any atom-count spectrum.

Proof. Let the atom cores be J_i, with $v_i = |V(J_i)|$, $e_i = |E(J_i)|$, and $r_i = e_i - v_i + 1$. Since a tree of k atoms makes $k - 1$ one-point identifications,

$$s = 1 + \sum_i (v_i - 1), \qquad \beta = \sum_i r_i. \tag{10.15}$$

If $\beta = 0$, every atom is one triple and (10.15) gives $k = s - 1$. If $\beta = 1$, there is one cyclic atom, whose core is an even cycle of order $v \geq 4$; the remaining atoms are single triples, so $s = k + v - 1$. This is exactly the middle line of (10.13).

Suppose $\beta \geq 2$. For positive integers a, b, the inequality $\sqrt{a} + \sqrt{b} \geq \sqrt{a+b-1} + 1$ follows after squaring from $(a-1)(b-1) \geq 0$. Hence $q(a) + q(b) \geq q(a+b-1) + 2 \geq q(a+b) + 1$, where the second inequality uses that q increases by at most one between consecutive integers. Together with Lemma 10.8, repeated application gives

$$s \geq 1 + k + q(\beta),$$

so $k \leq s-1-q(\beta)$. Conversely, for every integer in the displayed interval choose one cyclic core of rank β and order $s-k+1$, supplied by the lemma, and attach $k-1$ single-triple atoms. Equality in (10.14) forces one positive-rank atom and minimum order, proving the rigidity statement. □

The rigidity in the maximum concerns the distribution of cyclic rank: all positive rank lies in one minimum-order atom. It does not assert that the numerical parameters determine the graph-isomorphism type of that atom's core.

For $s \geq 3$, define

$$\alpha(s) = \begin{cases} s, & s \text{ even}, \\ s+1, & s \text{ odd}, \end{cases} \qquad \delta(s) = \left\lfloor \frac{(s-1)^2}{4} \right\rfloor + 1.$$

Theorem 10.10 (Indecomposability phase diagram)**.** *Apart from the single triple* $(|E(F)|, |V(F)|) = (1, 3)$*, a connected reduced one-point-indecomposable obligatory system with shadow order* s *and* m *hyperedges exists if and only if*

$$s \geq 4, \qquad \alpha(s) \leq m \leq \left\lfloor \frac{s^2}{4} \right\rfloor. \tag{10.16}$$

A connected reduced one-point-decomposable obligatory system exists if and only if

$$s \geq 3, \qquad s-1 \leq m \leq \delta(s). \tag{10.17}$$

Consequently, for $s \geq 3$*, the feasible connected parameters split into three exact zones:*

$$\begin{aligned} s-1 \leq m < \alpha(s) &\implies \textit{every system is decomposable}, \\ \alpha(s) \leq m \leq \delta(s) &\implies \textit{both behaviours occur}, \\ \delta(s) < m \leq \lfloor s^2/4 \rfloor &\implies \textit{every system is indecomposable}. \end{aligned} \tag{10.18}$$

For $m \geq 2$*, the last zone is equivalently*

$$s \leq \left\lceil 2\sqrt{m-1} \right\rceil. \tag{10.19}$$

Proof. By Corollary 5.4, the nontrivial indecomposable systems are J^+ with J finite, simple, bipartite and 2-connected. If $|V(J)| = s$ and $|E(J)| = m$, minimum degree two gives $m \geq s$; equality makes J a cycle and therefore forces s even. The upper bound is the usual $m \leq \lfloor s^2/4 \rfloor$. Even cycles, and for odd s a cycle on $s-1$ vertices plus one new vertex with two neighbours, give the lower endpoints; adding cross-edges fills the intervals.

A decomposable system admits a connected bipartite shadow with a cut vertex. If its bipartition sizes are $a+b=s$ and the cut vertex lies in the a-side, then deleting it gives components with sizes (a_i,b_i) and

$$m \le b + \sum_i a_i b_i \le b + (a-1)(b-1) = ab - a + 1.$$

The symmetric estimate gives $m \le ab - \min(a,b) + 1 \le \delta(s)$. The bound is sharp: take a connected bipartite graph on $s-1$ vertices with $m-1$ edges and adjoin one leaf. This also fills every value in (10.17). The three zones and the elementary inversion (10.19) follow. □

Corollary 10.11 (Rigidity on the structural boundaries)**.** *Let F be connected, reduced and obligatory, with shadow order s and $m = |E(F)|$ hyperedges.*

(1) If F is indecomposable, s is even and $m = s$, then $F \cong C_s^+$.
(2) If F is indecomposable, s is odd and $m = s+1$, then $F \cong \Theta(2a, 2b, 2c)^+$ for positive $a \le b \le c$ with $a+b+c = (s+1)/2$.
(3) If F is indecomposable and $m = \lfloor s^2/4 \rfloor$, then $F \cong K^+_{\lfloor s/2 \rfloor, \lceil s/2 \rceil}$.
(4) If F is decomposable and $m = \delta(s)$, then F is a one-point amalgamation of $K^+_{a,b}$ and one triple, where $a+b = s-1$ and $|a-b| \le 1$.

Here $\Theta(p,q,r)$ is the theta graph with three internally disjoint paths of lengths p, q, r and common endpoints.

Proof. The first and third assertions are the equality cases of minimum degree and the bipartite extremal bound. In the second case $\sum_v (\deg v - 2) = 2$. A single degree-four vertex would be a cut vertex, so there are exactly two degree-three vertices and all others have degree two; the three resulting paths have the same parity. Their total length is $s+1$, which is even, so all three paths are even. For the fourth assertion when $s \ge 5$, choose the shadow by assembling the canonical atom cores. Its cyclic blocks are precisely the nontrivial canonical cores. Equality in the cut-vertex estimate forces a balanced complete bipartite graph on $s-1$ vertices plus one leaf, and hence the displayed complete-bipartite atom and one singleton atom. The tree cases $s = 3, 4$ are immediate. □

Proposition 10.12 (Componentwise atom-count spectrum)**.** *Let F be reduced and obligatory with at least one hyperedge and $c \ge 1$ connected components, shadow order $s = |V(F)| - |E(F)|$, cycle rank $\beta = 2|E(F)| - |V(F)| + c$, and k canonical atoms. For every feasible (s, β, c), the possible values of k are exactly*

$$\begin{aligned} &\beta = 0: \quad k = s - c, \\ &\beta = 1: \quad c \le k \le s - c - 2, \quad k \equiv s - c \pmod 2, \\ &\beta \ge 2: \quad c \le k \le s - c - q(\beta). \end{aligned} \tag{10.20}$$

The parameters themselves are feasible exactly when $s \ge 2c$ for $\beta = 0$ and $s \ge 2c + q(\beta)$ for $\beta \ge 1$.

Proof. If the atom cores are J_i, a forest with k atom-nodes and c components makes $k-c$ identifications. Hence

$$s = c + \sum_i (|V(J_i)| - 1), \qquad \beta = \sum_i (|E(J_i)| - |V(J_i)| + 1).$$

The connected proof applies componentwise. Necessity gives the lower atom floor $k \geq c$ and the three upper formulas. For sufficiency, take $c-1$ single-triple components and realise the corresponding connected atom count in the remaining component. The feasibility thresholds follow by concentrating all positive rank in one minimum-order cyclic atom. □

Remark 10.13 (Blocks and the earlier reduction). Komjáth proved that a finite triple system is obligatory if and only if all of its 2-connected components are obligatory [18]. General hypergraph block decomposition and its relation to the incidence graph are classical; see Whitney [4], Diestel [5, Chapter 3], and, for the hypergraph formulation, Bahmanian and Šajna [30]. For the reduced 3-uniform systems considered here, the canonical atoms in Theorem 5.3 are precisely these maximal nonseparable edge-supported blocks. Thus the new content of the canonical atom theorem is not the existence of a block decomposition, but the exact identification of the block types that can occur in an obligatory triple system, together with the quantitative spectra derived from them.

Corollary 10.14 (Block formulation of the classification)**.** *Let F be a finite triple system. After isolated points are deleted, F is obligatory if and only if every nonempty block of F° is either one triple or J^+ for a finite 2-connected simple bipartite graph J.*

Proof. If $E(F^\circ) = \varnothing$, then F° is empty and the assertion follows from Lemma 1.1. Otherwise suppose first that F is obligatory. Lemma 1.1 gives that F° is obligatory, and Komjáth's reduction gives obligatoriness of each of its blocks. A block is one-point indecomposable, so Corollary 5.4 gives exactly the displayed list.

Conversely, each displayed block is obligatory by the positive expansion result (the one-triple case included). Komjáth's reduction therefore makes F° obligatory, and Lemma 1.1 restores the isolated points. □

Lemma 10.15 (Local product for one-point decompositions)**.** *Let F be reduced and obligatory with at least one hyperedge. Let $\mathcal{A}(F)$ be its canonical atoms, let $S(F)$ be the points belonging to at least two atoms, and put*

$$\mu(p) = |\{A \in \mathcal{A}(F) : p \in V(A)\}|.$$

Let $\mathcal{D}(F)$ be the set of partitions of $E(F)$ into nonempty connected supported pieces such that two pieces meet in at most one point and the piece–shared-point incidence graph is a forest, ordered by refinement. Then $\mathcal{D}(F)$ is a lattice and

$$\mathcal{D}(F) \cong \prod_{p \in S(F)} \Pi_{\mu(p)}, \tag{†}$$

where Π_r denotes the partition lattice of an r-element set. If F has k canonical atoms and c connected components, then

$$\operatorname{rk} \mathcal{D}(F) = k - c = \sum_{p \in S(F)} (\mu(p) - 1). \tag{‡}$$

The empty product in (†) *is the one-element lattice.*

Proof. A supported one-point forest decomposition cannot split a block: if two hyperedges of one block lay in different pieces, the unique path between those pieces in the piece–shared-point forest would contain a point whose deletion separates the two hyperedges, contradicting nonseparability of the block. Thus every element of $\mathcal{D}(F)$ is a coarsening of the canonical atom decomposition.

For a shared point p, restrict such a coarsening to the set of the $\mu(p)$ atoms through p; this gives an element of $\Pi_{\mu(p)}$. Conversely, choose independently a partition of the atoms through every shared point and take the equivalence relation on all atoms generated by these local identifications. To track its classes without contracting overlapping subtrees, replace the star at each shared point p by the following two-level tree. Retain a central node p, insert one node (p, B) for each block B of the prescribed local partition, join p to (p, B), and join (p, B) to each atom in B. Replacing a star of a forest by a tree with the same attachment nodes preserves acyclicity, so the resulting auxiliary graph is a forest.

Delete the edges from the central nodes p to their nodes (p, B). The components containing atoms are vertex-disjoint and correspond exactly to the generated equivalence classes; their atom unions are connected supported pieces. Two different blocks at one p cannot lie in the same component, since a path between them, together with their two edges to p, would give a cycle in the auxiliary graph. Contract these components in the full auxiliary graph, with the deleted edges restored, while retaining the central nodes p. There are therefore no repeated incidences between a piece and p. Delete each degree-one central node, which is now internal to one piece. The resulting graph is precisely the piece–shared-point incidence graph and is still a forest. In particular, two pieces cannot meet at two shared points. Restriction recovers the prescribed local partitions. Conversely, a connected supported piece is recovered from its local identifications, since its atom nodes are connected through the shared points it contains. Thus the constructions are inverse and preserve refinement, proving (†). For the related theory of connected set partitions and bond lattices, see Simon et al. [33] and Stanley [34, Lecture 2]. The factors here are full partition lattices on the atoms through each shared point, not the bond lattice of the incidence forest itself. Binary joints contribute two-element-chain factors, whereas three atoms through one point already give the diamond $\Pi_3 \cong M_3$, the familiar obstruction to distributivity [35, Chapter II, Theorem 101].

Finally the canonical incidence forest has $k + |S(F)|$ vertices, c components and $\sum_p \mu(p)$ edges. Hence

$$\sum_p (\mu(p) - 1) = k - c,$$

which is also the rank of the product lattice by the standard rank formula for partition lattices and their products [6, Chapter 3]. □

Lemma 10.16 (Capacity-safe realization of every attachment profile)**.** *Fix pairwise disjoint canonical atoms $A_1, \dots, A_k$ and an integer $1 \le c \le k$. For every integer partition*

$$\lambda = (\lambda_1, \dots, \lambda_t) \vdash k - c$$

(with the empty partition allowed when $k = c$), there is a forest assembly of the A_i into exactly c connected components whose shared-point excess profile is λ; that is, its nontrivial shared points have multiplicities $\lambda_i + 1$. Every atom is incident with at most two distinct shared points. Consequently, for a fixed atom list and fixed c, all these assemblies have the same order, size, shadow order and cycle rank.

Proof. Put $N = k - c$. If $N = 0$, take the $k = c$ atoms as separate components. Assume $N > 0$. Use $N + 1$ atoms in one nontrivial component and leave the remaining $c - 1$ atoms as singleton components.

If $t = 1$, identify one chosen vertex from each of the $N + 1 = \lambda_1 + 1$ active atoms to one common shared point. If $t \ge 2$, arrange shared points $p_1, \dots, p_t$ in a chain. Use $t - 1$ atoms as connectors, the ith connector meeting p_i and p_{i+1}. The two endpoint shared points already have one incident connector and the internal shared points have two. Supply the remaining incidences with leaf atoms so that $\deg(p_i) = \lambda_i + 1$. The number of required leaf atoms is

$$\lambda_1 + \lambda_t + \sum_{i=2}^{t-1} (\lambda_i - 1) = N - t + 2,$$

so together with the $t - 1$ connectors this uses exactly $N + 1$ atoms. Every connector atom uses two distinct vertices and every leaf atom one; canonical atoms have at least three vertices, so the identifications can be made with distinct ports inside each atom. The atom–shared-point incidence graph is a tree and has the required multiplicities.

Disjoint union and one-point amalgamation preserve obligatoriness by Lemma 4.1 and Proposition 4.4. Each shared point separates the incident atom branches in this forest assembly. A nonseparable supported block therefore cannot contain edges from two different atoms: the intervening shared point would separate those edges. Each supplied atom is itself one-point indecomposable, so its image is a maximal nonseparable supported block. By Theorem 5.3, these images are precisely the canonical atoms of the assembled system. Thus the constructed profile is the canonical attachment profile, not merely an auxiliary incidence pattern.

Any forest assembly of the fixed k atoms into c components makes exactly $k-c$ vertex identifications. Thus

$$|E(F)| = \sum_i |E(A_i)|, \qquad |V(F)| = \sum_i |V(A_i)| - (k-c),$$

so the stated global parameters do not depend on the chosen profile. □

Corollary 10.17 (Exact one-point factorization-lattice spectrum)**.** *Fix a feasible triple (s, β, c) for reduced obligatory systems with at least one hyperedge, where*

$$s = |V(F)| - |E(F)|, \qquad \beta = 2|E(F)| - |V(F)| + c, \qquad q(r) = \lceil 2\sqrt{r} \rceil .$$

As F ranges over all such systems with these fixed parameters, put k for its number of canonical atoms and $N = k - c$. The possible values of N are exactly

$$\begin{aligned} &\beta = 0: \quad N = s - 2c, \\ &\beta = 1: \quad 0 \le N \le s - 2c - 2, \quad N \equiv s - 2c \pmod 2, \\ &\beta \ge 2: \quad 0 \le N \le s - 2c - q(\beta). \end{aligned} \tag{$\star$}$$

For every allowed N, the possible lattice isomorphism types of $\mathcal{D}(F)$ are precisely

$$\prod_{i=1}^{t} \Pi_{\lambda_i + 1}, \qquad \lambda = (\lambda_1, \dots, \lambda_t) \vdash N, \tag{$\star\star$}$$

where for $N = 0$ the empty partition gives the one-element lattice. Every lattice in ($\star\star$) occurs, and different partitions λ give nonisomorphic lattices.

Proof. The rank spectrum ($\star$) is Proposition 10.12 after subtracting c from the atom count k. Fix an allowed N and choose one system with $k = N + c$ atoms and the prescribed global parameters. By Lemma 10.16, its fixed atom list can be reassembled with the same c and the same global parameters to realize every partition $\lambda \vdash N$. Lemma 10.15 then gives exactly the product in ($\star\star$). Conversely every system has an excess profile partitioning N, so there are no other types.

Finally, the standard characteristic polynomial of the partition lattice [6, Chapter 3] is

$$\chi_{\Pi_r}(x) = \prod_{j=1}^{r-1} (x - j),$$

and characteristic polynomials multiply under direct products, by the product rule for Möbius functions [6, 36]. For the equivalent braid-arrangement viewpoint, see Stanley [34, Lectures 1–2]. Hence the multiplicity of the root j in the characteristic polynomial of the lattice in ($\star\star$) is

$$|\{i : \lambda_i \ge j\}|.$$

These multiplicities recover the partition λ, so distinct partitions give nonisomorphic lattices. □

Uniqueness of indecomposable direct factors also has a classical incidence-poset formulation; compare Schmitt [37, Lemma 6.1]. The argument here recovers the attachment profile directly from characteristic-polynomial root multiplicities.

The numerical and lattice spectra record different parts of the same decomposition. The shadow order, cyclic rank and component count restrict the possible excess $N = k - c$. For each allowed N, the attachment profile $\lambda \vdash N$ determines the product of partition lattices, and every such profile is realizable. These assertions classify the available factorization-lattice types; they do not identify all systems with the same numerical parameters.

Cyclic costs and assembly freedom. The atom-count bound and the local product measure complementary constraints. The first records how much cyclic structure costs; the second records the decomposition choices left after the atoms have been fixed.

Proposition 10.18 (Exact cyclic-cost budget)**.** *Let F be connected, reduced and obligatory, with positive Levi cycle rank β. Put $s = |V(F)| - |E(F)|$, let k be its canonical atom count, and set $N = k - 1$. For the positive-rank atoms, write r_i for the core rank, v_i for the core order, and h for their number. Define*

$$C = \sum_{r_i>0} q(r_i) - q(\beta), \qquad L = \sum_{r_i>0} (v_i - 2 - q(r_i)).$$

Then

$$N + C + L = s - 2 - q(\beta), \qquad C \geq h - 1, \quad L \geq 0.$$

In particular,

$$N + (h - 1) + L \leq s - 2 - q(\beta).$$

The cost $C + L$ is zero exactly when the atom count is maximal: there is one minimum-order cyclic atom carrying rank β, and all other atoms are single triples.

Proof. Zero-rank atoms have core order two. Equation (10.15) therefore gives

$$s = 1 + k + \sum_{r_i>0} q(r_i) + L, \qquad \sum_{r_i>0} r_i = \beta.$$

Substituting the definitions of N and C gives the identity. Lemma 10.8 gives $L \geq 0$. Since $\beta > 0$, there is at least one positive-rank atom. The strict aggregation inequality in the proof of Theorem 10.9 gives $C \geq h - 1$. Thus zero cost forces $h = 1$ and zero core-order slack; the converse follows from the same identity. □

Since $N = \operatorname{rk} \mathcal{D}(F)$, the budget also describes the rank available to the factorization lattice. Splitting cyclic rank among atoms and increasing their core orders reduce this rank. It is a numerical constraint, not a graph-edit-distance stability theorem or a uniqueness statement about the cores.

We use the usual second-kind Stirling numbers: they count partitions into nonempty blocks, with $S(0, 0) = 1$ [7, Section 5.1].

Corollary 10.19 (Sharp counts at a prescribed piece number)**.** *Let F be reduced and obligatory with at least one hyperedge, k canonical atoms and c connected components. Put $N = k - c$ and let $\lambda = (\lambda_1, \dots, \lambda_t) \vdash N$ be its attachment excess profile. For $0 \le j \le N$, define*

$$d_j(F) = |\{D \in \mathcal{D}(F) : |D| = k - j\}|,$$

where $|D|$ counts the actual nonempty original-edge pieces. Write $S(a, b)$ for the number of partitions of an a-element set into b blocks. Then

$$\sum_{j=0}^{N} d_j(F) z^j = \prod_{i=1}^{t} \left(\sum_{r=0}^{\lambda_i} S(\lambda_i + 1, \lambda_i + 1 - r) z^r \right),$$

and

$$\binom{N}{j} \le d_j(F) \le S(N + 1, N + 1 - j) \qquad (0 \le j \le N).$$

If $N \ge 2$, then for every $1 \le j \le N - 1$ the lower equality holds exactly when all shared points are binary, that is, $\lambda = (1, \dots, 1)$; the upper equality holds exactly when $\lambda = (N)$. Each bound is attained at every j by its corresponding extremal profile, even with the atom list and component count fixed. For $N = 0$ the generating polynomial is 1, and for $N = 1$ the two extremal profiles coincide.

Proof. Let π_i be the local partition at the ith shared point. In the auxiliary forest from Lemma 10.15, delete the central point-nodes and their incident edges. The remaining forest has $k + \sum_i |\pi_i|$ vertices and $\sum_i (\lambda_i + 1)$ edges. Every component contains an atom and corresponds to one actual piece. Hence

$$|D| = k - \sum_i (\lambda_i + 1 - |\pi_i|).$$

Independent local choices give the generating polynomial. This is the standard partition-lattice enumeration [6, Example 3.10.4]; connected-partition polynomials and their articulation factorization are discussed in Simon et al. [33, Sections 3–5].

For the lower bound, give the ith local set one anchor and λ_i ordinary elements. A j-element subset of the total N ordinary elements defines local partitions by joining its elements to their respective anchors and leaving all others singleton. This is an injection into the choices counted by $d_j(F)$.

For the upper bound, identify all local anchors to one anchor on a set of $N + 1$ elements. Fuse the anchor blocks of the local partitions and retain every other block. Restriction to each anchored local set recovers its partition, so the map is injective. A tuple of total excess j gives $N + 1 - j$ blocks, proving the upper bound. This construction also shows that merging two positive profile parts never decreases any coefficient.

For strictness in the lower bound, suppose some $\lambda_i \ge 2$. Form an anchor-free pair in that group and choose $j - 1$ of the remaining $N - 2$ ordinary elements to join their local anchors. This tuple has excess j but is outside the lower image. For strictness in the upper bound, suppose there are at least two profile

groups. Choose $j+1$ ordinary elements meeting two groups, put them in one anchor-free block and leave all other elements singleton. This partition has $N+1-j$ blocks but is outside the upper image. Both choices are possible for $1 \le j \le N-1$. The two extremal profiles attain the respective bounds. Finally, Lemma 10.16 realizes both profiles using the same atoms and component count, with all global parameters preserved. □

For example, take C_4^+ and two single triples. A chain assembly with two binary joints has decomposition polynomial $1+2z+z^2$; an assembly with both triples attached at one point of C_4^+ has polynomial $1+3z+z^2$. Both systems have twelve vertices, six hyperedges, three canonical atoms and cycle rank one, and both maximize the atom count for these parameters. Their four and five decomposition choices differ only because of attachment geometry. Thus extremal cyclic concentration does not eliminate assembly freedom. The counting inequalities above are elementary partition-lattice consequences; their role here is to quantify that freedom, not to assert a new general enumeration theorem. The individual factors are reversed rows of second-kind Stirling numbers, whose generating-function and concavity properties are classical [38]. The bounds above instead compare different attachment profiles at fixed total excess. Pitman's probabilistic representations and coefficient bounds give further context for these classical Stirling rows [39]. Our question is which products of the rows are realized by attachment profiles, not a new general identity for Stirling numbers.

11. Quantitative comparisons for graph cores

The results in this section concern the graph J underlying an expansion J^+, not uncountably chromatic hypergraph hosts. If J is a simple bipartite subdivision of K_4, its six path lengths $L_{ij} \ge 1$ satisfy $L_{ij} \equiv \varepsilon_i + \varepsilon_j \pmod 2$ for some branch colours $\varepsilon_i \in \{0,1\}$. Its path interiors are mutually disjoint and $|V(J)| = |E(J)| - 2$. The canonical-atom theorem admits every such J^+; it does not supply the comparisons below.

Exact edge laws and their entropy cost. Let Ω be a finite nonempty alphabet, let κ be a positive probability law on it, and let λ be a symmetric pair law with both marginals κ. Write $R_G = \kappa^{V(G)}$ and

$$D(\nu\|R) = \sum_x \nu(x) \log \frac{\nu(x)}{R(x)}, \qquad I(\lambda) = D(\lambda\|\kappa^{\otimes 2}),$$

with natural logarithms and $0 \log 0 = 0$. An exact edge witness on G has singleton law κ at every actual vertex and pair law λ on every oriented actual edge. Prescribing an averaged edge law is a different problem. The relation between entropy maximization and graph density inequalities is developed in Szegedy [9]; a bound for one scalar density is not substituted here for exact marginal constraints.

Theorem 11.1 (Potts edge witnesses)**.** *Let G be a finite simple graph with m edges, each of which has an endpoint of degree at most three. Let $q \geq 2$, let κ be uniform on $\{1, \dots, q\}$, and put*

$$\lambda_s(x,y) = \begin{cases} s/q, & x = y, \\ (1-s)/(q(q-1)), & x \neq y, \end{cases} \qquad 0 \leq s \leq 1.$$

If $s < 1/q$, assume additionally that G is bipartite. There is an exact edge witness τ with

$$D(\tau \| R_G) \leq m I_q(s), \qquad I_q(s) = s \log(qs) + (1-s) \log \frac{q(1-s)}{q-1}.$$

Disconnected graphs and isolated vertices are included.

The theorem includes every bipartite K_4 subdivision and every uniform Potts pair law. Its antiferromagnetic part retains bipartiteness; it does not extend to arbitrary pair tables by permutation symmetry. The proof uses a constrained finite Gibbs fit. Two-colour conditioning supplies the correlation sign, following the mechanism in Csikvári and Lin [8]; a conditional comparison at a low-degree endpoint makes every zero-coupling edge fit. A reverse-KL interpolation gives the sharp cost. These steps, including zero-support endpoints, are detailed in Appendix B.

Corollary 11.2 (Strict slack and full-table perturbations)**.** *Under the preceding graph hypotheses, suppose G has a cycle and $0 < s < 1$, $s \neq 1/q$. The fitted Potts witness has $D(\tau \| R_G) < m I_q(s)$. There is a neighbourhood of (κ, λ_s), relative to compatible positive singleton and symmetric pair laws, in which every input admits an exact witness with $D(\tau \| \kappa^{V(G)}) < m I(\lambda)$. For $q \geq 3$ the neighbourhood may be chosen inside the class of pairs having a positive exchangeable triple extension.*

This is a full-table neighbourhood: it includes nonuniform singleton laws and non-Potts tables. Its radius is not asserted to be uniform, and no openness is inferred at $s = 0, 1$ or at independence. At independence a separate argument is available.

Theorem 11.3 (Exact tables near independence)**.** *Fix a finite simple bipartite graph G containing a cycle, with girth g, and a positive law κ on an alphabet of size $q \geq 2$. For a symmetric pair law λ with marginal κ, set*

$$f(x,y) = \frac{\lambda(x,y)}{\kappa(x)\kappa(y)} - 1, \qquad C_{xy} = \sqrt{\kappa(x)\kappa(y)}\, f(x,y), \qquad \rho = \|C\|_{\mathrm{op}}.$$

There is a positive, explicitly specified radius $\delta(G, \kappa)$ such that $0 < \rho \leq \delta$ implies the existence of a full-support exact witness with

$$D(\tau \| \kappa^{V(G)}) \leq |E(G)| I(\lambda) - \rho^g/8.$$

At $\rho = 0$ the independent law works. For a forest, an exact witness satisfies $D(\tau \| \kappa^{V(G)}) = |E(G)| I(\lambda)$ by a tree construction, without a smallness condition.

The local signed-density framework is already established [10]. Here a nearly fitted pairwise Gibbs seed is corrected in all unary and edge-table coordinates. The positive shortest even-cycle term pays for that correction; the proof and its conservative constants are in Appendix B. Small scalar density error alone would not supply the prescribed marginals.

Proposition 11.4 (Cycle surplus for disjoint alphabet blocks)**.** *Let G have n vertices, m edges and c components, and write $\beta = m - n + c$. Partition the alphabet into nonempty blocks with masses $\pi_i > 0$. In block i, let κ_i and λ_i be compatible singleton and symmetric pair laws. Suppose each connected component C has an exact within-block witness of cost at most $|E(C)|D(\lambda_i\|\kappa_i^{\otimes 2})$. Set $\kappa = \sum_i \pi_i\kappa_i$ and $\lambda = \sum_i \pi_i\lambda_i$, with the pair law supported on same-block pairs. There is an exact witness τ_0 with*

$$D(\tau_0\|\kappa^{V(G)}) \le mI(\lambda) - \beta H(\pi).$$

If a known exact witness has slack Δ, mixing its pair and joint laws with ε times their independent reference laws gives a new exact witness with slack at least $(1-\varepsilon)\Delta - mh(\varepsilon)$, where h is binary entropy.

Proof. Choose one block independently in each connected component, then use its within-block witness. Disjoint supports give the exact component cost $(|V(C)|-1)H(\pi) + \sum_i \pi_i D(\tau_{i,C}\|\kappa_i^{V(C)})$. The pair cost is $I(\lambda) = H(\pi) + \sum_i \pi_i I(\lambda_i)$. Summing proves the first bound, including isolated vertices. For the mixture statement, joint KL convexity bounds the new joint cost by $(1-\varepsilon)$ times the old cost. The pair-level chain rule gives

$$(1-\varepsilon)I(\lambda_0) = I((1-\varepsilon)\lambda_0 + \varepsilon\kappa^{\otimes 2}) + I(B; X, Y),$$

where B is the Bernoulli mixture label and $I(B; X, Y) \le h(\varepsilon)$. Subtracting gives the claimed slack. □

The term $\beta H(\pi)$ permits a global block choice to cost entropy while retaining a sharp edge budget. It cannot be recovered by imposing a separate sharp bound on every projected pair law. For example, on the six-vertex bipartite K_4 subdivision with path lengths $(2,1,1,1,1,2)$ in the order ab, ac, ad, bc, bd, cd, take uniform κ on three symbols and

$$\lambda = \frac{1}{5400}\begin{pmatrix} 1787 & 7 & 6 \\ 7 & 599 & 1194 \\ 6 & 1194 & 600 \end{pmatrix}. \tag{1}$$

This positive table has a positive exchangeable triple extension. An explicit exact witness satisfies $8I(\lambda) - D(\tau\|\kappa^6) > 229/540$. Its projected $\{1,2\}$-indicator pair is independent, but its global indicators need not be; the full budget pays for them. The construction and all positivity checks are given in Appendix B. This covers the displayed table on the displayed graph, not arbitrary path lengths or all tables. The graph $K_{3,3} - e$ already lies in the scalar-density class of Conlon et al. [12]. Combined with the universal entropy formulation in Szegedy [9, Corollary 9.1], this already supplies the sharp exact-pair entropy bound on

this graph for every pair law. The example records an explicit budget-transfer construction, not a new graph or table family.

Problem 11.5 (The unrestricted finite-alphabet question)**.** Suppose λ is the pair marginal of an exchangeable law on Ω^3. Does every actual simple bipartite K_4 subdivision J admit an exact witness satisfying $D(\tau\|\kappa^{V(J)}) \le |E(J)|I(\lambda)$? Zero entries in the pair and triple laws are allowed.

The results above settle specified families, not this problem. Exact feasibility itself is easy: broadcast one pair sampled from λ to the two bipartition classes. This may cost too much entropy. Even for binary data, gluing stationary Markov bridges can impose incompatible branch correlations. Thus the remaining issue is a global sharp budget, not merely an inverse fit or translation into Lean.

Powers of one weighted regular host. Let X be finite, let $\mu_x > 0$ with $\sum_x \mu_x = 1$, and let $W : X^2 \to [0,1]$ be symmetric with $\sum_y \mu_y W(x,y) = p > 0$. Composition and inner products use this original measure. Set $T = W/p$, $A = T^2$, $K_j = A^j$, $\Pi(x,y) = 1$ and $I(x,y) = \mathbf{1}_{x=y}/\mu_y$. For six positive integers a_e, indexed by the edges of K_4, put

$$F(a) = \int_{X^4} \prod_{e=vw} K_{a_e}(x_v, x_w)\, d\mu^4.$$

If $J(2a)$ replaces edge e by a path of length $2a_e$, then

$$F(a) = \frac{t(J(2a), W)}{p^{2\sum_e a_e}},$$

where t denotes weighted homomorphism density. This reduction concerns all-even subdivisions; it does not cover every bipartite parity pattern.

Theorem 11.6 (Two centered spectral levels)**.** *Suppose, on the complete space orthogonal to constants, A has at most two eigenvalues. Equivalently,*

$$A = yI + (1-y)\Pi + (x-y)P, \qquad 0 \le y \le x \le 1,$$

where P is an orthogonal projection with $P1 = 0$. Then $F(a) \ge 1$ for every choice of the six positive integers a_e. Both spectral multiplicities are unrestricted.

The whole centered spectrum is meant: two nonzero eigenvalues together with an additional zero complement give three levels and are not covered. The proof places every power in the cone generated by I, Π and one common nonnegative positive-semidefinite Markov kernel. Weighted identity contractions reduce the cone expansion to a finite list of quotients of K_4; their comparisons are proved in the appendix. Single-kernel positive-spectrum inequalities are studied by Sidorenko [11]. Neither that literature nor the cone argument permits replacing one common host by six unrelated PSD kernels.

The appendix also shows strictness unless $A = \Pi$ when $|X| \ge 2$. For fixed powers this gives a neighbourhood of actual roots that may have more than two spectral levels. More precisely, for symmetric nonnegative Markov roots

T, T_0 on the same (X, μ), write $N = \sum_e a_e$, $M_0 = \|T_0\|_\infty$ and $\varepsilon = \|T - T_0\|_\infty$, using maximum absolute kernel entries. Markov averaging and telescoping give

$$|F(a;T) - F(a;T_0)| \le 2N\varepsilon(M_0 + \varepsilon)^5.$$

Indeed, $\|T^j - T_0^j\|_\infty \le j\varepsilon$ and every positive power is bounded by $M_0 + \varepsilon$; telescope the six factors. This elementary stability observation is host- and tuple-dependent, not uniform compensation for arbitrary interacting spectra.

Problem 11.7 (The unrestricted common-power question)**.** For the same actual weighted regular host, is

$$\int_{X^4} K_k(ab)K_{r+h}(ac)K_u(ad)K_r(bc)K_u(bd)K_\ell(cd)\, d\mu^4 \ge 1$$

whenever $k \ge u \ge 1$, $r \ge \ell \ge 1$ and $h \ge 1$, without further spectral or mixing hypotheses?

This question remains open here. Individual centered terms and individual spectral contributions can be negative even for actual Markov-square hosts; the full density retains cycle, diamond and constant-mode surplus. An arbitrary PSD operator without a nonnegative Markov square root is not an admissible counterexample. These restrictions also explain why the two open problems cannot be closed by a scalar calculation for a single test host. The subdivision results of Im et al. [13, Theorems 1.3, 1.4 and 1.6] provide important known cases: uniform even-theta substitution, nonuniform substitution under divisibility conditions, and a clique subdivision with one exceptional vertex. We do not treat these cases as new results or infer arbitrary nonuniform path lengths from them.

Collective compensation in signed projection lifts. The two-level theorem constrains the full spectrum. The next result instead constrains the geometry of one signed channel, while retaining arbitrary coarse spectra and all six positive exponents. Its point is not positivity of each centered contribution: some cubic contributions are negative even in the class below. The proof absorbs that error into the quadratic surplus under the original probability law.

Proposition 11.8 (Weighted flat-complement transfer)**.** *Let (I, π) have $n \ge 4$ points and positive weights summing to one. Let S be a symmetric nonnegative Markov kernel relative to π. Choose $f_i \in \{-1, 1\}$ and put*

$$Q(i,j) = \frac{\mathbf{1}_{i=j}}{\pi_j} - f_i f_j, \qquad \delta = \max_i |\pi_i^{-1} - n|.$$

Suppose

$$\delta < n - 2, \qquad \delta^2(n + \delta - 1) \le 8(n-3)(n-2-\delta).$$

Let H be a self-adjoint kernel with $H^2 = \beta Q$, $\beta \ge 0$, and $|H(i,j)| \le S(i,j)$. On the original law $\mu(i,\varepsilon) = \pi_i/2$, $\varepsilon \in \{-1, 1\}$, define

$$T((i,\varepsilon),(j,\eta)) = S(i,j) + \varepsilon\eta H(i,j).$$

This is an actual nonnegative self-adjoint Markov root. For every six positive integers a_e*,*

$$F(a;T) \geq F(a;S) + (n-1) \sum_{C \in \mathcal{C}(K_4)} \beta^{\sum_{e \in C} a_e},$$

where $\mathcal{C}(K_4)$ *consists of its four triangles and three quadrilaterals. Thus* $F(a;S) \geq 1$ *implies* $F(a;T) \geq 1$*, strictly if* $\beta > 0$*. If a lift in this class had* $F(a;T) < 1$*, its coarse root would also violate the inequality; for* $\beta > 0$ *its density would be strictly smaller.*

The restriction on π is independent of S and of the exponents; $\delta \leq 1$ suffices in every dimension $n \geq 4$. The signs need not be centered or be eigenfunctions of S. The projection has rank $n-1$, and neither balance of the smoothed source trace nor a small source norm is assumed. For example, the construction in the source-bound research supplement has thirteen coarse states, an unbalanced rank-twelve channel, and fourteen distinct values on the entire centered space of the fine square, including zero. The proposition is a structured comparison, not a reduction of arbitrary roots to signed lifts. Its proof and a stronger spectral remainder are in Appendix B. The separate finite-cover certificates in that research record concern one host and connected covering degree at most three; arbitrary simultaneous covers remain open. No such certificate is used to infer the unrestricted common-power inequality.

Scope of formal verification

The classification and specified finite structural extensions have verified Lean 4 implementations [40, 41]. The accepted results include the canonical atoms, finite parameter and connected atom-count spectra, classical-block identification, local decomposition products, actual-piece accounting and supported-cover increments. They use the original carriers and retain their exact finiteness and structural hypotheses.

Verification of the extended manuscript is not complete. In particular, none of the entropy–density statements in Section 11 has an accepted Lean implementation. Moreover, fixed-atom profile realization, the full lattice spectra, maximal-chain and height endpoints, the complete phase/equality interfaces and the new coefficient bounds remain distinct obligations. The arithmetic deficit theorem supports the cyclic-budget calculation, but does not by itself identify every remaining geometric or lattice endpoint. The proofs in the main text establish the manuscript conclusions; successful typechecking of other statements cannot replace them. Implementation details, exact exported types, validation chronology and the remaining crosswalk are separated into Appendix A.

Appendix A. Formalization interfaces and reproducibility

The implementation uses Lean 4 [40] and the mathematical library Mathlib [41]. These references describe the proof assistant and library, not the verification

status of this manuscript. Formal verification checks a precisely stated theorem; its correspondence with a prose formulation still requires mathematical review [42]. Verified typechecking addresses a distinct trust question [43]; no use of that external checker is claimed here.

The self-contained Lean 4 source is available in the public `Erdos593` repository. The repository contains a classification implementation in commit `6fd00a7`, whose Git timestamp is 15 July 2026. A dated public validation record is the successful GitHub Actions run of 19 July 2026, at commit `88ce701`. A Git timestamp alone does not establish the date of public disclosure or priority. Li's arXiv v2, submitted on 23 July 2026, reports a separate formal verification of the broader results in his paper. These records do not date either author's private work, and no shared code or derivation between the two Lean developments is asserted. For every finite triple system F, the exported theorems establish

$$\texttt{F.IsObligatory} \iff \texttt{F.isolatedReduction.Intrinsic},$$

$$\texttt{F.IsObligatory} \iff \texttt{Constructible F.isolatedReduction}.$$

These exported theorems verify both directions of the classification, including the isolated-vertex reduction. Here "finite" describes the system F being classified. The host triple systems are not assumed finite; they are quantified within the documented ambient-universe convention of the formalisation.

Further accepted Lean results cover the canonical-atom normal form and minimal-generator classification, the finite parameter spectra, the connected atom-count spectrum and its maximisers, the unrestricted classical-block equivalence, the forward atomic cycle and even-theta boundary forms, and the supported decomposition order isomorphism with local partition products. A separately validated extension has passed canonical and standalone replay and final source reconciliation for the actual-piece/local-partition block-count identity; these results retain their stated finiteness, structural and feasible-parameter hypotheses. The block-count checkpoint is now included in the synchronized public and private repositories. The abstract capacity-two incidence-forest construction has also been validated, but it does not yet formalise assembly of an arbitrary fixed list of actual atoms.

The supported-subtype cover criterion, the unique local coordinate changed by a cover, and the additive piece-deficit identity have now passed pinned canonical and standalone Lean replay and final live source reconciliation. This extension also verifies that coarsening along a cover reduces the actual piece count by one and increases the deficit by one. These results use actual original-edge pieces and local quotient carriers. The extension is included in the public repository through the reviewed cover/grading integration. Maximal-chain lengths and height endpoints remain separate obligations.

Verification of the extended manuscript is therefore partial. Remaining interfaces include the maximal-chain grading and height endpoints, fixed-atom profile realization and canonical-label fidelity after assembly, the componentwise atom-count and full factorization-lattice spectra, the all-maximal-block running-order

statement, and sorted theta parameter normalization. The full indecomposability phase diagram, decomposable spectrum and decomposable equality classification also remain separate obligations. The characteristic-polynomial recovery of the attachment profile is not claimed as an additional Lean endpoint. Exact statement-by-statement coverage is tracked in the manuscript crosswalk.

The accepted atom-deficit theorem supplies the arithmetic identity and positive-atom bound behind Proposition 10.18, under its exact structural, positive-rank and additive-count hypotheses. Its identification with lattice rank still uses the manuscript's grading argument. The generating polynomial, coefficient bounds and equality cases in Corollary 10.19 are proved here but are not yet additional Lean endpoints. Counting by actual pieces uses the accepted product and block-count interfaces, not an assumption about covers in an ambient partition lattice; the fixed-atom sharpness assertion retains the separate assembly-realization obligation.

The Lean development does not reproduce every intermediate prose lemma verbatim. In particular, its negative direction uses a shift-graph host in place of the paper's high-odd-girth input, and its closure construction uses a singular-safe layering rather than the displayed cofinality-indexed chain. These are distinct proof routes to the classification. Kernel validation supplements rather than replaces mathematical review; a full external audit of the extended development remains pending.

Appendix B. Exact marginal entropy and common-power densities

All logarithms in this appendix are natural. For a strictly positive probability law R on a finite set, we use $D(\nu\|R) = \sum_x \nu(x)\log(\nu(x)/R(x))$, with $0\log 0 = 0$. Every marginal below refers to a specified vertex or edge, not an average over edges. We prove the restricted statements in Theorems 11.1, 11.3 and 11.6. The arbitrary-table entropy question and the unrestricted common-power inequality remain open.

Uniform Potts marginals.

Proof of Theorem 11.1. Write $G = (V, E)$, $m = |E|$, and let R be the uniform product law on $\{1, \dots, q\}^V$. The one-symbol case is immediate, so assume $q \geq 2$. For $0 < s < 1$, put

$$s_0 = \frac{1}{q}, \qquad \theta(s) = \log\frac{(q-1)s}{1-s}, \qquad I_q(s) = s\log(qs) + (1-s)\log\frac{q(1-s)}{q-1}.$$

For an interaction vector $J = (J_e)_{e\in E}$, define

$$Z(J) = \mathbb{E}_R \exp\left(\sum_{uv\in E} J_{uv}\mathbf{1}_{\{X_u = X_v\}}\right),$$

$$\frac{d\nu_J}{dR} = Z(J)^{-1}\exp\left(\sum_{uv\in E} J_{uv}\mathbf{1}_{\{X_u = X_v\}}\right).$$

Minimize $\log Z(J) - s\sum_e J_e$ on $[0,\theta(s)]^E$ if $s > s_0$, and on $[\theta(s),0]^E$ if $s < s_0$. A minimizer exists and has full support. Simultaneous permutations of the q colours preserve its law. Thus each singleton is uniform and every ordered pair is determined by its agreement probability: all diagonal entries are equal, as are all off-diagonal entries. It remains to fit every edge agreement to s.

The removed-edge sign. For nonnegative interactions, any two distinct vertices have agreement probability at least $1/q$. For nonpositive interactions on a bipartite graph, vertices in opposite parts have agreement probability at most $1/q$. Here is a finite proof of both assertions. Condition on the set of vertices whose colours lie in $\{1,2\}$ and on every colour outside this set. An edge from the set to its complement always has unequal colours and introduces no field. The remaining law is a zero-field Ising model with couplings $J_e/2$. For nonnegative couplings, the high-temperature expansion expresses its correlation numerator as a sum over edge sets with boundary $\{u,v\}$, with nonnegative products of $\tanh(J_e/2)$; its denominator is a positive sum over even edge sets. The correlation is therefore nonnegative. For nonpositive interactions on a bipartite graph, reversing the spins on one part gives the preceding nonnegative-coupling model, and reverses the correlation between opposite parts. Averaging the conditioning compares the probabilities of colours $(1,1)$ and $(1,2)$. By colour symmetry these are respectively a/q and $(1-a)/(q(q-1))$, where a is the agreement probability. Their comparison gives the asserted sign. This is the weighted two-colour conditioning mechanism used in the proper-colouring setting by Csikvári and Lin [8].

Nonzero interactions fit. The derivative in coordinate $e = uv$ is $a_e - s$, where $a_e = \nu_J(X_u = X_v)$. An interior coordinate therefore fits. At the far face $J_e = \theta(s)$, remove this interaction and call the resulting agreement probability a_0. Restoring it multiplies agreement odds by $\exp\theta(s)$, so the new agreement is

$$\frac{e^{\theta(s)}a_0}{1+(e^{\theta(s)}-1)a_0}.$$

The removed-edge sign shows that this is at least s in the positive regime and at most s in the negative regime. At the upper face the derivative is nonpositive; at the lower face it is nonnegative. These Karush–Kuhn–Tucker inequalities give the reverse comparison in each regime. Hence every nonzero interaction fits, including a far-face one.

Zero interactions fit. Suppose $J_{vw} = 0$, and choose the endpoint v with degree at most three. There are at most two other neighbours with nonzero interactions. If these are a, b, write their interactions as J_1, J_2. Conditioning on all colours except X_v gives

$$\nu_J(X_v = X_a) - \nu_J(X_v = X_b) = \nu_J(X_a \neq X_b)\frac{e^{J_1}-e^{J_2}}{q-2+e^{J_1}+e^{J_2}}.$$

The left side is zero because both nonzero incident interactions fit. Full support makes $\nu_J(X_a \neq X_b) > 0$, so $J_1 = J_2$. For a positive common interaction, the

colour of either active neighbour maximizes the conditional probability of X_v, whether the two neighbour colours agree or differ. Consequently the agreement on vw is at most that on va, namely s. The lower-face derivative inequality gives agreement at least s. For a negative common interaction, neighbour colours minimize the conditional probability and both comparisons reverse. With one active neighbour the same maximum or minimum comparison applies. With none, agreement on vw is $1/q$, contrary to the relevant zero-face inequality when $s \neq s_0$. Thus every original edge fits exactly. Colour symmetry now gives the whole prescribed Potts table, not just agreement. Isolated vertices remain independent uniform variables.

The entropy bound. For each interior u on the segment from s_0 to s, choose a fitted law ν_u as above, and take $\nu_{s_0} = R$. Set $C_u = D(\nu_u \| R)$ and $S_u = \sum_e J_e(u)$. Direct substitution gives

$$D(\nu_v \| \nu_u) = C_v - C_u + (u - v)S_u \geq 0. \tag{2}$$

If $s_0 \leq v < u \leq s$, then $C_u - C_v \leq m(u - v)\theta(u)$. If $s \leq u < v \leq s_0$, then $S_u \geq m\theta(u)$ and the negative factor $u - v$ again gives $C_u - C_v \leq m(u - v)\theta(u)$. Fix the target law ν_s, telescope these inequalities along a finite mesh from s_0 to s, and let its mesh tend to zero. The oriented Riemann sums yield

$$C_s \leq m \int_{s_0}^{s} \theta(u)\, du = m I_q(s).$$

This step requires neither continuity nor uniqueness of the chosen fitted family. In particular, the order of the two laws in (2) is essential in the negative regime.

At $s = s_0$, use R. At $s = 0$ or $s = 1$, apply the interior result to $s_\varepsilon = (1 - \varepsilon)s + \varepsilon/q$ and take a convergent subsequence in the finite joint simplex. Marginalization and divergence relative to the positive reference R are continuous, including at zero joint masses. The limit gives all endpoint marginals and the bound. The assertion is vacuous on edges when E is empty, and the product law then has zero cost. □

Cyclic slack and perturbation of the full pair table.

Proof of Corollary 11.2. Keep the notation of the preceding proof, with $0 < s < 1$, $s \neq s_0$, and with at least one cycle. First, every minimizing interaction is nonzero. Indeed, at a hypothetical zero edge the preceding neighbour-colour comparison is strict on the positive-probability event $X_a = X_b \neq X_w$ when there are two active neighbours, or $X_a \neq X_w$ when there is one. Simplicity ensures that these are distinct actual vertices. Averaging contradicts the relevant zero-face inequality. The case of no active neighbour was already excluded.

If $e = uv$ lies on a cycle, removing its interaction leaves an alternative u–v path with nonzero interactions. In the two-colour conditioning, every conditional correlation has the weak sign established above. The event that all vertices have colours in $\{1, 2\}$ has positive probability. On this event the Ising high-temperature numerator contains the strictly positive contribution of the alternative path, after a bipartition spin reversal in the negative regime. The

removed-edge agreement is therefore strictly above $1/q$ in the positive regime and strictly below $1/q$ in the negative regime. Reinsertion at the interaction cap would strictly overshoot s, contradicting the cap inequality. Hence

$$0 < J_e(s) < \theta(s) \quad (s > s_0), \qquad \theta(s) < J_e(s) < 0 \quad (s < s_0)$$

on every cycle edge. Bridges need not be strictly inside their cap.

The covariance matrix of the statistics $h_e = \mathbf{1}_{\{X_u = X_v\}}$ is positive definite under every full-support law. To see this, restrict a putatively constant linear combination to two colours and write $h_e = (1+\sigma_u\sigma_v)/2$. The distinct simple-edge Walsh monomials are linearly independent modulo constants, so every coefficient must vanish. Thus $\log Z$ has positive-definite Hessian and an injective gradient. The analytic implicit function theorem supplies a unique locally analytic fitted vector $J(s)$; uniqueness patches these local inverses along the applicable interval. That interval is $0 < s < 1$ for a bipartite graph, and $s_0 \le s < 1$ in the general positive regime. No negative-regime fit is asserted for a nonbipartite graph. The zero vector is the unique fit at s_0.

For the exact fit,

$$C(s) = s \sum_e J_e(s) - \log Z(J(s)), \qquad C'(s) = \sum_e J_e(s).$$

The terms involving $J'(s)$ cancel because every edge agreement is s. The cap bounds imply $J(s) \to 0$ at s_0, and hence

$$\Delta_s := mI_q(s) - C(s) = \int_{s_0}^{s} \left(m\theta(u) - \sum_e J_e(u) \right) du > 0. \tag{3}$$

Above s_0 the integrand is positive because a cycle edge is strictly inside its cap. Below s_0 the integrand is negative and the orientation of the integral reverses. This proves strictness in both regimes.

To perturb the whole table, fix a mean-zero orthonormal basis $f_1, \dots, f_{q-1}$ for the uniform singleton law κ_0. Use every unary statistic $f_i(X_v)$ and, for a fixed orientation of each edge, every statistic $f_i(X_u)f_j(X_v)$. These statistics are orthonormal under κ_0^V, since distinct supports leave an unmatched centred factor. They are therefore affinely independent modulo constants on the joint configuration space, and their covariance under the full-support law ν_s is positive definite. The expectation map for exponential tilts of ν_s is locally invertible. It fits every sufficiently nearby compatible singleton and symmetric pair table: the unary and pair-product basis reconstructs every full actual-edge table. The basis is fixed while the target singleton law varies. The tilted joint law depends continuously on these moments. Both $D(\tau\|\kappa^V)$ and $mD(\lambda\|\kappa^{\otimes 2})$ are continuous for positive nearby inputs. By (3), their difference remains negative after shrinking the neighbourhood. This gives an open family of non-Potts tables and nonuniform singleton laws, but not coverage of the entire compatible-table domain.

For $q \ge 3$, this neighbourhood can also remain within the positive exchangeable-triple cone. Let η_{iid} be the uniform product triple, η_{dist} the

uniform ordered all-distinct triple, and η_{eq} the uniform all-equal triple. A positive extension of the base table is

$$\eta_s = qs\eta_{\mathrm{iid}} + (1 - qs)\eta_{\mathrm{dist}} \quad (0 < s < s_0),$$

and

$$\eta_s = \frac{1-s}{1-s_0}\eta_{\mathrm{iid}} + \left(1 - \frac{1-s}{1-s_0}\right)\eta_{\mathrm{eq}} \quad (s_0 < s < 1).$$

For a nearby symmetric probability-table perturbation $\lambda' = \lambda_s + \delta$, set $d_x = \sum_y \delta(x, y)$ and define

$$\begin{aligned} H(x,y,z) = {} & \delta(x,y)\kappa_0(z) + \delta(x,z)\kappa_0(y) + \delta(y,z)\kappa_0(x) \\ & - d_x\kappa_0(y)\kappa_0(z) - d_y\kappa_0(x)\kappa_0(z) - d_z\kappa_0(x)\kappa_0(y). \end{aligned}$$

This is symmetric, has total mass zero, and its (x, y) marginal is $\delta(x, y)$: the two row-drift terms cancel and $\sum_z d_z = 0$. Thus $\eta_s + H$ extends λ' and has singleton $\kappa_0 + d$. It stays strictly positive for sufficiently small perturbations. This last assertion is deliberately restricted to $q \geq 3$; the binary triple-extension cone has additional boundary restrictions. □

An explicit neighbourhood of independence.

Proof of Theorem 11.3. Let G be finite, simple and bipartite, let $n = |V|$, $m = |E|$, and let κ be a positive law on a q-symbol alphabet, $q \geq 2$. For a symmetric table λ with marginals κ, put

$$a = \min_x \kappa(x), \qquad f(x,y) = \frac{\lambda(x,y)}{\kappa(x)\kappa(y)} - 1,$$
$$C_{xy} = \sqrt{\kappa(x)}\, f(x,y) \sqrt{\kappa(y)}, \qquad \rho = \|C\|_{\mathrm{op}}.$$

If G is a forest, root each component, draw its root from κ, and draw each child using transition $\lambda(x, y)/\kappa(x)$, independently conditional on its parent. This has the required marginals and divergence $mD(\lambda\|\kappa^{\otimes 2})$ by the chain rule, including zero pair entries. Isolated vertices contribute zero.

Suppose now that G has even girth $g \geq 4$. The following constants specify the radius used in the theorem:

$$d = n(q-1) + m(q-1)^2, \quad A = 2^m a^{-m}, \quad M = a^{-1},$$
$$B = \sqrt{d}\,(M+1)A, \quad L = \sqrt{d}\,M, \quad r = \frac{\log 2}{4L},$$
$$\delta_0 = \min\Big\{1, \frac{a}{3}, \frac{1}{2A}, \frac{a\log 2}{8m}, \left(\frac{r}{8B}\right)^{1/(g-1)}, \left(\frac{1}{8B^2}\right)^{1/(g-2)}\Big\}.$$

Assume $0 < \rho \leq \delta_0$. Every matrix entry of C has magnitude at most ρ, so $|f(x, y)| \leq \rho/a \leq 1/3$. With $R = \kappa^V$, take the full-support seed

$$w(X) = \prod_{uv \in E} (1 + f(X_u, X_v)), \qquad Z = \mathbb{E}_R w, \qquad \nu_0 = Rw/Z.$$

This seed is not assumed to have the target marginals.

Expand Z over actual edge subsets. A subset with a leaf contributes zero, because each κ-weighted row sum of f is zero. A surviving nonempty subset has at least g edges, and a surviving g-edge subset is exactly a shortest cycle. Its contribution is $\operatorname{Tr}(C^g)$, which is at least ρ^g because g is even and C is real symmetric. The absolute total of all terms with at least $g+1$ edges is at most $A\rho^{g+1}$: a k-edge term is bounded by $(\rho/a)^k$, and there are at most 2^m terms. Similarly $|Z-1| \le A\rho^g$. Therefore

$$Z \ge 1 + \rho^g - A\rho^{g+1} \ge 1 + \frac{\rho^g}{2}, \qquad \log Z \ge \frac{\rho^g}{4}. \tag{4}$$

The last inequality uses $\rho \le 1$. The even-cycle mechanism is the finite weighted version of the local signed-density argument in Lovász [10]. We still need to correct the seed to exact marginals; a scalar partition bound alone does not do so.

Choose a κ-orthonormal mean-zero basis $\xi_1, \dots, \xi_{q-1}$. Let $T = (T_1, \dots, T_d)$ contain all unary statistics $\xi_i(X_v)$ and all oriented actual-edge products $\xi_i(X_u)\xi_j(X_v)$. Under R they are mean zero and orthonormal, including on distinct edges sharing a vertex. Each has magnitude at most M, since $|\xi_i(x)| \le 1/\sqrt{\kappa(x)}$. The seed potential has the complete gauge decomposition

$$h_0 = \sum_{uv \in E} \log(1 + f(X_u, X_v)) = \theta_0 \cdot T + c.$$

Let b be the desired moment vector: its unary coordinates vanish and its edge coordinates are $\sum_{xy} \lambda(x,y)\xi_i(x)\xi_j(y)$. Each of the latter has magnitude at most ρ, by the definition of C. Matching b fixes every singleton and every full edge table, because $\{1, \xi_1, \dots, \xi_{q-1}\}$ is a basis of all functions on the alphabet.

We bound the seed residual explicitly. In a unary-inserted edge-subset expansion, every unmarked leaf kills the term. A nonempty forest has at least two leaves, so every surviving term contains a cycle and at least g edges. For an edge statistic marked at adjacent u, v, a surviving forest can only be a single u–v path. The one-edge path contributes exactly the desired coordinate of b; every other path has at least $g-1$ edges, as it closes with uv to a cycle. Cyclic terms have at least g edges. Thus each unnormalized edge moment is $b_i + E_i$ with $|E_i| \le MA\rho^{g-1}$. Division by $Z \ge 1$ adds at most $|b_i||Z-1| \le A\rho^{g+1}$. The unary bound is no larger. Consequently

$$\|\mathbb{E}_{\nu_0} T - b\|_2 \le B\rho^{g-1}. \tag{5}$$

Consider the convex function

$$F(\theta) = \log \mathbb{E}_R e^{\theta \cdot T} - \theta \cdot b.$$

Its gradient at θ_0 is the residual in (5), and its Hessian is the covariance of T under its Gibbs law. The range of h_0 is at most $4m\rho/a \le (\log 2)/2$, using $|\log(1+f)| \le 2|f|$. A displacement of Euclidean norm at most r adds potential

range at most $2Lr = (\log 2)/2$. Throughout the closed radius-r ball about θ_0, the Gibbs density relative to R is therefore at least $1/2$. For every $z \in \mathbb{R}^d$,

$$\mathrm{Var}_{\nu_\theta}(z \cdot T) = \inf_c \mathbb{E}_{\nu_\theta}(z \cdot T - c)^2 \geq \frac{1}{2} \inf_c \mathbb{E}_R(z \cdot T - c)^2 = \frac{1}{2}\|z\|_2^2.$$

Hence $\nabla^2 F \geq \frac{1}{2} I$ on that ball. Write $G_0 = \|\nabla F(\theta_0)\|_2 \leq B\rho^{g-1} \leq r/8$. Integration along a segment gives

$$F(\theta_0 + z) \geq F(\theta_0) + \nabla F(\theta_0) \cdot z + \frac{1}{4}\|z\|_2^2.$$

On the radius-r boundary this exceeds $F(\theta_0)$ by at least $r^2/8$. A minimum on the closed ball is therefore interior, has zero gradient, and is a global minimum by convexity. Its Gibbs law τ has full support and fits b exactly. The same inequality gives $F(\theta_0) - F(\theta_*) \leq G_0^2$.

Put $I = D(\lambda \| \kappa^{\otimes 2})$. The target expectation of h_0 is mI, so the gauge constant cancels to give $F(\theta_0) = \log Z - mI$. Since the fitted law has moment vector b, $D(\tau \| R) = -F(\theta_*)$. Combining these facts yields

$$\begin{aligned} D(\tau \| R) &\leq mI - \frac{\rho^g}{4} + B^2 \rho^{2g-2} \\ &\leq mI - \frac{\rho^g}{8}, \end{aligned}$$

where the final step uses $\rho^{g-2} \leq 1/(8B^2)$. At $\rho = 0$, the table is $\kappa^{\otimes 2}$ and the product joint works. If a triple extension is required, it is supplied in this smallness regime by the exchangeable probability law

$$\eta(x, y, z) = \kappa(x)\kappa(y)\kappa(z)[1 + f(x, y) + f(x, z) + f(y, z)].$$

Its bracket is nonnegative, its total mass is one, and its pair table is λ. No triple-extension hypothesis was used to construct τ. The radius depends on G, q and the smallest singleton mass; the argument does not cover arbitrary tables far from independence. □

The three-colour budget-transfer example. Here we verify the assertion following Proposition 11.4 without a numerical entropy estimate. Let $G = K_{3,3} - e$, with the actual subdivision and table in (1), and let R be uniform on its $3^6 = 729$ colourings. Put

$$\lambda_0 = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/9 & 2/9 \\ 0 & 2/9 & 1/9 \end{pmatrix}.$$

Its blocks have masses $1/3, 2/3$. The second block has uniform binary singletons and agreement $1/3$. The Potts proof supplies a Gibbs witness τ_B with interactions in $[-\log 2, 0]$, hence every one of its 2^6 configurations has probability at least 2^{-14}. Choose all colours equal to 1 with probability $1/3$, and choose τ_B on $\{2, 3\}$ with probability $2/3$; call the resulting law τ_0. The block identity gives slack at least $3H(1/3, 2/3)$. Set $\nu = (99/100)\tau_0 + (1/100)R$ and

$\lambda_b = (99/100)\lambda_0 + (1/100)\kappa^{\otimes 2}$. Then $8I(\lambda_b) - D(\nu\|R) \geq (99/100)3H(1/3, 2/3) - 8h(1/100)$.

Let $D_{12} = D_{21} = 1/5400$, $D_{11} = D_{22} = -1/5400$, with other entries zero. Its row sums vanish and $\lambda = \lambda_b + D$. For $\psi = (1, -1, 0)$, define

$$g(x, y) = \frac{D_{xy}}{\kappa_x \kappa_y} = -\frac{\psi(x)\psi(y)}{600}, \qquad \tau(X) = \nu(X) + R(X) \sum_{uv \in E(G)} g(X_u, X_v).$$

Every singleton correction vanishes. At any actual edge, the direct correction is D and every other edge correction vanishes after integrating a centred endpoint. Thus total mass, singletons and every full edge table are exact. If a colouring contains colour 1 and another colour, its nontrivial vertex cut has at least two edges because G is bridgeless. Consequently $\sum_{uv} \psi(X_u)\psi(X_v) \leq 6$, so the negative correction is paid by $R/100$. With no colour 1, the extra lower bound $(99/100)(2/3)2^{-14} = 33/819200 > 1/218700$ pays the possible deficit $R(8/600 - 1/100)$. The all-1 colouring has mass at least $33/100$ before correction and is also nonnegative afterwards. These cases exhaust all configurations, so τ is an actual probability law.

The pair correction has total variation $d_P = 1/2700$, and joint variation is at most $d_V = 2/675$. Maximal coupling gives $|H(P) - H(Q)| \leq h(d) + d\log(N-1)$ on N states; the same bound controls KL changes relative to a uniform law. Its right side is increasing for $0 \leq d \leq 1 - 1/N$. Both d_P and the upper bound d_V lie in their respective intervals, so substitution of the joint upper bound is valid. Applying it to $N = 9,729$ therefore gives slack at least

$$\frac{99}{100}3H(1/3, 2/3) - 8h(1/100) - 8[h(d_P) + d_P \log 8] - [h(d_V) + d_V \log 728].$$

Use $H(1/3, 2/3) > \log(3/2) > 1/3$ and $h(t) \leq t\log(1/t) + t$. The elementary bound $e > 8/3$ implies $\log 100 < 5$, $\log 8 < 3$, $\log 2700 < 9$, $\log 728 < 8$ and $\log(675/2) < 7$. The slack is strictly greater than

$$\frac{99}{100} - \frac{48}{100} - \frac{104}{2700} - \frac{32}{675} = \frac{229}{540}.$$

For triple feasibility, let η_0 put mass $1/3$ on 111 and mass $1/3$ uniformly on each permutation orbit 223 and 233. Its pair is λ_0. The exchangeable law

$$\eta(x, y, z) = \frac{99}{100}\eta_0(x, y, z) + \frac{1}{100}\kappa_x\kappa_y\kappa_z + \frac{D_{xy} + D_{xz} + D_{yz}}{3}$$

has pair λ and is positive: its IID contribution is $1/2700$ and its correction is no smaller than $-1/5400$. Finally, the projected $\{1, 2\}$ indicator has pair probability $4/9$ for two ones and singleton probability $2/3$, hence an independent pair. The construction makes no demand that its joint indicators be independent.

A common-kernel cone on four vertices. Let (X, μ) be a nonempty finite probability space with $\mu_x > 0$. Kernels act by $(Sf)(x) = \sum_y \mu_y S(x, y) f(y)$; all compositions, traces and inner products below refer to this original measure.

Define $I(x,y) = \mathbf{1}_{\{x=y\}}/\mu_y$ and $\Pi(x,y) = 1$. An operator power S^j is not an entrywise power.

Lemma B.1 (Common-kernel cone)**.** *Let S be one symmetric, entrywise nonnegative, positive-semidefinite Markov kernel on (X,μ). For each of the six edges of K_4, choose L_e independently in* $\operatorname{conv}\{I,\Pi,S\}$*. Then*

$$\mathbb{E}_{\mu^4} \prod_{uv\in E(K_4)} L_{uv}(X_u,X_v) \geq 1.$$

The same actual kernel S is used at all six edges.

Proof. Since $S1 = 1$ and S is positive semidefinite, $S-\Pi$ is positive semidefinite. Thus $S(x,x) \geq 1$. Every eigenvalue of S is nonnegative and the constant eigenvalue is one, so $\operatorname{Tr}(S^j) \geq 1$ for every positive integer j. The entrywise square $R(x,y) = S(x,y)^2$ is positive semidefinite as well. Indeed, with $D = \operatorname{diag}(\mu)$ and $B = D^{1/2}SD^{1/2}$, its weighted matrix is $D^{-1/2}(B\circ B)D^{-1/2}$; apply the Schur product theorem. We use that the trace of a product of two positive semidefinite operators is nonnegative, but never assume a corresponding sign for a product of three.

Expand the six convex combinations. Their 3^6 elementary assignments of I,S,Π have nonnegative coefficients summing to one. Fix one such assignment. In its I-edge graph choose a spanning forest with the same components. On the equality support of that forest, each deleted cyclic I edge is $1/\mu_x \geq 1$; off its support the forest product is zero. All other factors are nonnegative, so deleting these cyclic edges lowers the density. Contract the remaining I trees. A tree with t vertices contributes $\mu_x^t \mu_x^{-(t-1)} = \mu_x$. Every quotient vertex therefore retains the original measure μ. Delete Π factors and the resulting S loops, which are at least one. Parallel S edges must be retained. We now check all quotient types.

Four quotient vertices. There was no contraction. The retained simple graph is a subgraph of K_4. Ordinary leaves integrate to one using $S1 = 1$. Its leaf-stripped core is empty, a triangle, a four-cycle, a diamond, or K_4. The first three have densities 1, $\operatorname{Tr}(S^3)$, and $\operatorname{Tr}(S^4)$ respectively. Writing S^2 for operator composition, the diamond density is

$$\mathbb{E}_{c,d} S(c,d)[S^2(c,d)]^2 \geq [\mathbb{E}_{c,d} S(c,d)S^2(c,d)]^2 = [\operatorname{Tr}(S^3)]^2 \geq 1,$$

by Jensen under the probability measure $\mu_c\mu_d S(c,d)$. For K_4, set $f_{cd}(x) = S(x,c)S(x,d)$. Its density is

$$\mathbb{E}_{c,d} S(c,d)\langle f_{cd}, Sf_{cd}\rangle \geq \mathbb{E}_{c,d} S(c,d)(\mathbb{E}_x f_{cd}(x))^2,$$

because $S-\Pi$ is positive semidefinite and $S(c,d) \geq 0$. The last expression is the diamond density. This finite argument also makes explicit the complete-graph positive-kernel bound discussed in Sidorenko [11].

Three quotient vertices. The original partition sizes are $(2,1,1)$; write its edge multiplicities as $(s,t,v) \in \{0,1,2\}^2 \times \{0,1\}$, where the two first edges meet the

merged vertex. This gives all eighteen possibilities, including the empty graph. If $v = 0$, the density is

$$\mathbb{E}_a \left(\mathbb{E}_b S(a,b)^s\right) \left(\mathbb{E}_c S(a,c)^t\right) \geq 1,$$

because each bracket is one for exponent zero or one and at least one for exponent two by rowwise Jensen. If $v = 1$ and an arm is absent, integrating an ordinary leaf leaves at most a double edge, bounded by Jensen. The remaining possibilities are $(1,1,1)$, $(2,1,1)$, $(1,2,1)$ and $(2,2,1)$. The first is a triangle. Using the entrywise square R, either middle case has density

$$\operatorname{Tr}(RS^2) \geq \operatorname{Tr}(R\Pi) = \mathbb{E}_{x,y} S(x,y)^2 \geq 1,$$

since R and $S^2 - \Pi$ are positive semidefinite. The last case has density

$$\operatorname{Tr}(SR^2) \geq \operatorname{Tr}(\Pi R^2) = \|R1\|^2_{L^2(\mu)} \geq [\mathbb{E}_{x,y} S(x,y)^2]^2 \geq 1.$$

Here R^2 is operator composition, not a fourth entrywise power.

Two or one quotient vertices. Two vertices have partition sizes $(3,1)$ or $(2,2)$, hence at most four parallel edges. Their density is $\mathbb{E}S(x,y)^j \geq 1$ by Jensen for $j \geq 1$; for $j = 0$ it is one. One quotient vertex has only the already deleted loops and also has density one. Every elementary assignment is therefore bounded below by one, proving the convex-mixture assertion. Zeros in S, unequal weights, reducibility, and repeated eigenvalues cause no divisions or changes of measure. □

Proof of Theorem 11.6. Let P be an orthogonal projection on $L^2(\mu)$ with $P1 = 0$, and assume that the actual nonnegative kernel is

$$A = yI + (1-y)\Pi + (x-y)P, \qquad 0 \leq y \leq x \leq 1.$$

The mutually orthogonal projections $\Pi, P, I - \Pi - P$ give

$$A^j = y^j I + (1-y^j)\Pi + (x^j - y^j)P \qquad (j \geq 1).$$

If $x > y$ and $y < 1$, put $\alpha = (x-y)/(1-y)$ and $S = \Pi + \alpha P$. This is symmetric, positive semidefinite and Markov. For distinct u, v, $S(u,v) = A(u,v)/(1-y) \geq 0$, while $S(u,u) = 1 + \alpha P(u,u) \geq 1$ because P is positive semidefinite. Thus S is an actual nonnegative kernel, not merely a formal spectral projection. Write

$$a_j = y^j, \qquad b_j = \frac{(1-y)(x^j - y^j)}{x-y}, \qquad c_j = 1 - a_j - b_j.$$

Then $A^j = a_j I + b_j S + c_j \Pi$, and the coefficients are nonnegative:

$$\frac{x^j - y^j}{x-y} = \sum_{i=0}^{j-1} x^{j-1-i} y^i \leq \sum_{i=0}^{j-1} y^i = \frac{1-y^j}{1-y}.$$

Apply Lemma B.1 to the six positive exponents, independently. No ordering of them and no rank-one restriction on P is needed.

If $x = y < 1$, use $A^j = y^j I + (1-y^j)\Pi$ directly. If $y = 1$, then $x = 1$ and $A = I$; its six-edge density is $\sum_z \mu_z^{-2} \geq 1$. If $y = 0$, the preceding argument

includes the zero complementary band, and when $x = 0$ the kernel is simply Π. These cases avoid every potentially zero denominator. For $|X| \geq 2$, equality holds precisely when $A = \Pi$. When $y > 0$, the all-I assignment has positive weight and density $\sum_z \mu_z^{-2} > 1$. When $y = 0$ and $A \neq \Pi$, the all-S assignment with $S = A$ has positive weight, and

$$t(K_4, A) \geq [\mathrm{Tr}(A^3)]^2 = [1 + \mathrm{rank}(P)x^3]^2 > 1.$$

All remaining elementary terms are at least one.

The representation is a condition on the *complete* centred spectrum: when $x, y > 0$, an additional centred zero eigenspace is not included. The same-base-kernel condition of the lemma is also essential; the proof does not assert a bound for unrelated, even commuting, nonnegative positive-semidefinite kernels. Nor does this restricted density result give a universal exact-edge entropy witness; the universal entropy formulation and its graph-density relation require their full quantifiers [9]. □

Weighted projection transfer.

Proof of Proposition 11.8. All coarse kernels and compressed traces use $L^2(\pi)$; the lifted root acts on $L^2(\mu)$. Since $\mathbb{E}_\pi f^2 = 1$, $Q = I - f \otimes f$ is an orthogonal projection of rank $d = n - 1$. Put $B = S^2$, $E = \operatorname{ran} Q$, and define the actual compressed multiplication fields

$$R_i = QD_{\mathbf{1}_i/\pi_i}Q|_E, \qquad M_s(i) = (B^s R)_i, \qquad X_s(i) = M_s(i) - I_E.$$

Then $\mathbb{E}_\pi R_i = I_E$, and each $M_s(i)$ is positive semidefinite. For a complete real orthonormal eigenbasis $\phi_0 = 1, \phi_j$ of B, write $B\phi_j = \lambda_j \phi_j$, $0 \leq \lambda_j \leq 1$, and $Z_j = QD_{\phi_j}Q|_E$. Every mode except the chosen constant is retained. In particular, additional stationary modes are not discarded. Set $a_i = \pi_i^{-1} - n$ and $\sigma_t = \mathrm{tr}(B^t) - 1 = \sum_{j>0} \lambda_j^t$. For centered modes, expansion of the two and three projection factors gives

$$\mathrm{tr}(Z_j Z_k) = (n-2)\mathbf{1}_{j=k} + \mathbb{E}_\pi(a\phi_j\phi_k),$$
$$\mathrm{tr}(Z_j Z_k Z_l) = (n-3)\mathbb{E}_\pi(\phi_j\phi_k\phi_l) + \mathbb{E}_\pi(a\phi_j\phi_k\phi_l).$$

Indeed, $\operatorname{tr} D_g = \sum_i g_i = \mathbb{E}_\pi[(n+a)g]$, whereas $\mathrm{tr}((f \otimes f)D_g) = \mathbb{E}_\pi g$. The remaining terms contain an individual centered mean and vanish. Hence, with $b_0 = n - 2 - \delta > 0$,

$$g_{st} := \mathbb{E}_\pi \operatorname{tr}(X_s X_t) = \sum_{j>0} \lambda_j^{s+t} \|Z_j\|_{\mathrm{HS}}^2 \geq b_0 \sigma_{s+t}.$$

The trilinear trace of real symmetric matrices is permutation invariant, so order $x \leq y \leq z$. Write $L_s = B^s - \Pi$ and let $\mathcal{H} = L_x \circ L_y \circ L_z$ be the entrywise product. The Schur product theorem makes $\mathcal{H}$ positive semidefinite under the original weighted law. The triple identity above gives

$$C := \mathbb{E}_\pi \operatorname{tr}(X_x X_y X_z) = (n-3)\langle 1, \mathcal{H}1\rangle + \langle a, \mathcal{H}1\rangle \geq -\frac{\langle a, \mathcal{H}a\rangle}{4(n-3)}.$$

The inequality is completion of the square in the same positive quadratic form, not a sign assertion for a product of three matrices. Since $0 \preceq L_x \preceq I - \Pi$, its kernel entries have absolute value at most $M = n + \delta - 1$. Thus weighted Cauchy–Schwarz gives

$$0 \le \langle a, \mathcal{H}a \rangle \le \delta^2 M \, \mathbb{E}_{\pi^2} |L_y L_z| \le \delta^2 M \sqrt{\sigma_{2y}\sigma_{2z}}.$$

Linear terms vanish because $\mathbb{E}_\pi X_s = 0$. Spectral ordering and arithmetic–geometric mean now imply

$$\begin{aligned} J_{xyz} - d &:= \mathbb{E}_\pi \operatorname{tr}(M_x M_y M_z) - d \\ &\ge b_0 \sigma_{x+z} + b_1(\sigma_{x+y} + \sigma_{y+z}) \ge 0, \\ b_1 &= b_0 - \frac{\delta^2 M}{8(n-3)} \ge 0. \end{aligned}$$

Here $\sigma_{2y} \le \sigma_{x+y}$ and $\sigma_{2z} \le \sigma_{y+z}$. In particular $J_{xyz} = d$ forces $B = \Pi$, hence $S = \Pi$ by self-adjointness; the converse is immediate. To check the simpler sufficient range $\delta \le 1$, note that $\delta \le \sqrt{n}/2$, $b_0 \ge (n-2)/2$, $M \le 5n/4$, and $(n-3)(n-2) \ge n^2/8$. Therefore $\delta^2 M \le 5n^2/16 < 4(n-3)(n-2) \le 8(n-3)b_0$.

It remains to transfer this estimate to the full density. The entrywise condition makes T nonnegative, and averaging the middle sign with probability $1/2$ proves both its Markov property and

$$T^{2s}((i,\varepsilon),(j,\eta)) = S^{2s}(i,j) + \varepsilon\eta\beta^s Q(i,j).$$

Expanding the six factors and averaging the four independent signs leaves exactly the empty subset and the seven cycles of K_4. The empty contribution is $F(a;S)$. The rank-one identity $\operatorname{tr}(R_a R_b R_c) = Q(a,b)Q(b,c)Q(c,a)$ identifies the triangle coefficient. Its complementary three-edge star has some exponents x, y, z, and its contraction is exactly J_{xyz}.

For a quadrilateral, let L, M be its two complementary coarse powers. Both are nonnegative PSD Markov kernels. Set $v_{b,d}(a) = Q(a,b)Q(a,d)$ and $G(a,c) = \mathbb{E}_{b,d} M(b,d) v_{b,d}(a) v_{b,d}(c)$. This is a nonnegative mixture of positive rank-one operators. The quadrilateral contraction is therefore

$$\langle L, G \rangle_{\mathrm{HS}} \ge \langle \Pi, G \rangle_{\mathrm{HS}} = \langle M, Q \circ Q \rangle_{\mathrm{HS}} \ge \langle \Pi, Q \circ Q \rangle_{\mathrm{HS}} = d.$$

The first inequality uses $L - \Pi \succeq 0$, the equality uses $Q^2 = Q$, and the second inequality uses the PSD Schur square. Combining all seven terms proves the proposition. More precisely, each triangle contributes the additional nonnegative remainder $\beta^{\sum_{e \in C} a_e}[b_0\sigma_{x+z} + b_1(\sigma_{x+y} + \sigma_{y+z})]$. If $\beta = 0$, self-adjointness gives $H = 0$ and $F(a;T) = F(a;S)$; if $\beta > 0$, the cycle margin is strictly positive. □

Acknowledgments

The author thanks Tom de Groot for his advice on revising the manuscript for greater clarity. The author received no funding for this work and declares no competing interests.

AI assistance

OpenAI's GPT-5.6 Pro through ChatGPT and Aristotle [44] were used in proof development, adversarial checking, editorial restructuring, and the Lean formalisation. In particular, ChatGPT reviews informed the checking and presentation of the local forest-counting and capacity arguments in Section 10; Aristotle supplied the abstract incidence-forest construction distinguished from fixed-atom realization in the verification section. The author takes full responsibility for the arguments, citations, and final manuscript; newly incorporated material in this review draft remains subject to the author's review.

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